
CAUSALMIX: Defining and Benchmarking Hybrid Causal Discovery

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Abstract

1 Classical causal discovery recovers structure from numerical data, while large
2 language models (LLMs) carry semantic priors about variables and mechanisms
3 from pretraining. Research that combines the two is nascent and lacks a unified
4 problem definition or controlled evaluation protocol: existing benchmarks cover
5 the data-only and semantic-only endpoints but leave the middle ground untested.
6 To bridge the gap, in this work, we make two contributions. First, we formalize
7 *hybrid causal discovery* and release CAUSALMIX, a simulation and evaluation suite
8 that generates matched observational and interventional datasets from Bayesian-
9 network graphs and turns the regime into controlled attribution tests by varying
10 semantic cues, data budget, and data representation while fixing the graph. The
11 suite reports a semantic-only reference (what variable names alone recover) and a
12 data-only reference (what classical algorithms recover from the same samples), so
13 any gain from a hybrid method can be attributed to its inputs. Across ten LLMs on
14 four reference graphs, our controlled evaluation shows that zero-shot prompting
15 recovers substantial structure from names alone, while adding numerical evidence
16 yields weak and format-sensitive gains that fall well short of classical baselines on
17 the same data. Second, as a preliminary step beyond this baseline, we introduce
18 a post-training procedure that uses ground-truth graphs as verifiers to generate
19 observational and interventional supervision for LLM fine-tuning. The fine-tuned
20 model shows early evidence of combining semantic priors with numerical data,
21 with consistent gains on several small graphs, although a gap to classical methods
22 remains. We release CAUSALMIX as an open-source toolkit to support a more
23 rigorous evaluation protocol for LLM-assisted causal discovery.

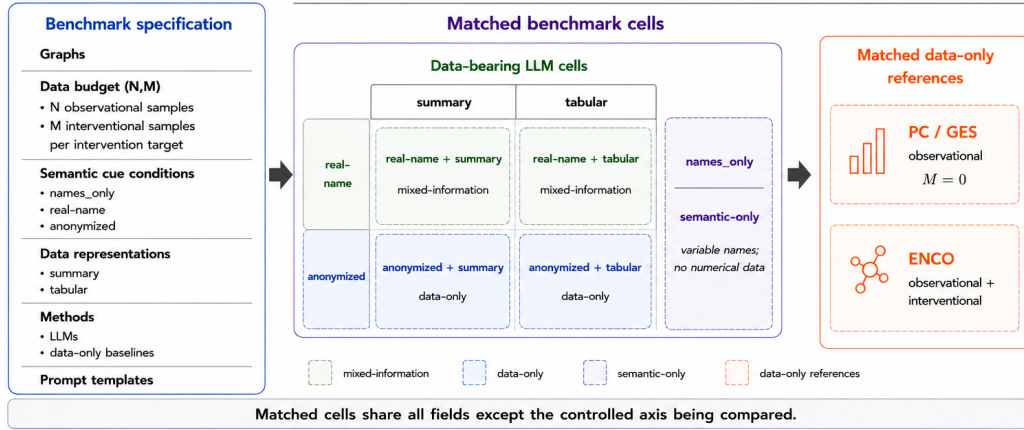
1 Introduction

24 Causal discovery in scientific domains rarely starts from numbers alone. Practitioners know what
25 the variables mean, which mechanisms are plausible, and which relations would contradict domain
26 knowledge; at the same time, observed and post-intervention samples are the empirical evidence
27 needed to separate causal structure from plausible stories. Classical causal-discovery methods exploit
28 the numerical evidence but ignore variable meanings, while LLM-based methods exploit names,
29 descriptions, and domain knowledge but may recover familiar benchmark structure from semantic
30 priors rather than from the supplied data. When such a method outputs a plausible graph, a headline
31 score alone does not reveal whether the method used semantics, numerical evidence, or both. We call
32 this setting *hybrid causal discovery* (HYBRID-CD): the learner is given semantic side information
33 (variable names, descriptions, domain text) and numerical evidence (observational and optionally
34 interventional samples), and must recover a causal graph from either or both.

35 The regime matters because many proposed uses of LLMs in causal discovery are motivated precisely
36 by the hope that semantic knowledge and statistical evidence can complement each other. It is hard
37

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(a) One Specification Generates Matched Cells



(b) Matched-cell contrasts

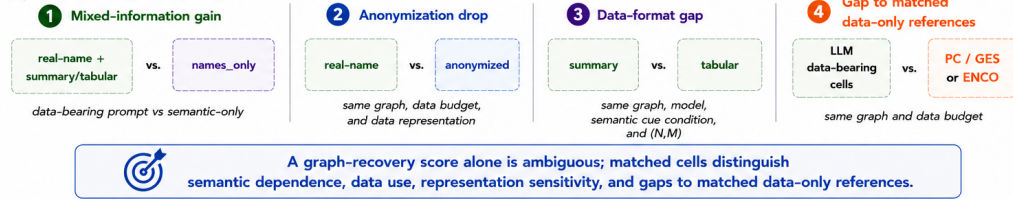


Figure 1: Overview of CAUSALMIX. A benchmark specification expands graphs, data budgets, representations, models, and baselines into matched evaluation cells. Each cell fixes the graph and method while varying one axis (semantic cue, data budget, or data representation), enabling attribution of recovery scores to information sources rather than to confounded experimental setups.

because the two channels can support the same edge for different reasons, and standard graph-recovery metrics do not attribute performance to the information source that produced it. Thus the scientific question is not only whether a method scores high, but whether it improves beyond what variable names alone provide and whether that improvement survives when names are anonymized.

Recent LLM-assisted methods target this middle ground by adding statistical summaries to prompts, decomposing graph construction into LLM queries, or combining language-model knowledge with data-driven evidence [1, 16, 37, 26]. However, they are evaluated under different graphs, sample counts, prompt formats, variable-name conditions, and classical baselines, so reported gains cannot be attributed to inputs: they may come from memorized variable names, the supplied data, the data representation, or their interaction. The field lacks a controlled protocol that compares semantic-only LLM prompting, data-only classical baselines, and hybrid methods on the same graph and data. Concretely, such a protocol must isolate four levers—a names_only prompt that estimates what an LLM recovers from variable semantics with no data, original-name and anonymized-name prompts at the same data budget that test dependence on names, two data representations (summary and tabular) that probe format sensitivity, and classical methods run on the same samples that bound what the data alone support—and report all four alongside any headline score.

Contributions. This paper makes two contributions to LLM-assisted causal discovery:

- **A simulation and evaluation suite for HYBRID-CD.** We formalize HYBRID-CD and release CAUSALMIX, an open-source benchmark suite that generates observational and interventional samples from Bayesian-network graphs and assembles matched evaluation cells. Each cell fixes the graph and varies one axis (semantic cue, data budget, data representation, or method class), while a names_only LLM prompt and a classical data-only baseline (PC [34], GES [10], ENCO [20]) bracket every hybrid run. Applied to Cancer, Earthquake, Asia, and Sachs across ten LLMs, the suite shows that names alone already account for most apparent recovery and that adding data yields weak, format-sensitive gains that remain far below the matched classical baselines—thus establishing a falsifiable target for future HYBRID-CD methods.

64 • **A post-training procedure with preliminary evidence of combining semantic priors with**
 65 **data.** As a first step beyond zero-shot prompting, we fine-tune Qwen3-4B with a verifier-based
 66 curriculum that uses ground-truth graphs to grade observational and interventional completions.
 67 Under the same controls, the fine-tuned model improves both parse validity and graph-recovery
 68 scores on several small graphs over the zero-shot baseline, providing early evidence that targeted
 69 training can push LLMs toward genuinely combining semantic priors with numerical evidence,
 70 although a gap to classical methods remains on harder graphs.

71 **Empirical highlights.** Across Cancer, Earthquake, Asia, and Sachs, CAUSALMIX reveals substantial
 72 recovery from variable names alone—exact recovery in 7 of 15 displayed cells on the three smaller
 73 graphs and nontrivial names-only F1 on Sachs. At the primary Sachs budget ($N=5000$, $M=200$),
 74 prompts that include data move relative to names_only by only -0.04 to $+0.23$ directed-edge F1
 75 for observational evidence and -0.11 to $+0.20$ F1 for interventional evidence, and anonymizing
 76 variable names reduces F1 in 28 of 36 paired comparisons. As the interventional budget grows,
 77 the GPT-5.2-Pro real-name summary prompt remains near $F1 = 0.56$ – 0.57 , while ENCO [20]
 78 reaches $F1 = 0.97$ – 1.00 at the same budget. CAUSALMIX therefore makes explicit a falsifiable
 79 target for future LLM-assisted causal discovery: improve over names_only, remain robust under
 80 anonymization, and move toward the matched classical reference under the same graph and data
 81 conditions.

82 2 Problem Formulation and Related Benchmarks

83 We formalize *hybrid causal discovery* (HYBRID-CD): causal graph recovery when the learner has
 84 access to both semantic side information and numerical observational or interventional evidence. The
 85 formulation makes explicit the attribution problem that hybrid and LLM-assisted pipelines implicitly
 86 raise but rarely test.

87 **Setup.** Let $X_1, \dots, X_d$ be random variables and $G = (\mathcal{V}, \mathcal{E})$ a directed acyclic graph (DAG) on
 88 $\mathcal{V} = \{1, \dots, d\}$ with adjacency matrix $A \in \{0, 1\}^{d \times d}$ [27]. A HYBRID-CD method takes *semantic*
 89 *side information* $S = \{s_i\}_{i=1}^d$ associated with each X_i (e.g., names or descriptions), *observational*
 90 *data* $\mathcal{D}_{\text{obs}} = \{x^{(k)}\}_{k=1}^N$, and optional *interventional data* $\mathcal{D}_{\text{int}} = \{\mathcal{D}_{\text{int}}^{(i)}\}_{i \in \mathcal{I}}$ over intervention targets
 91 $\mathcal{I} \subseteq \mathcal{V}$, and returns an estimate $\hat{A} = \mathcal{A}(S, \mathcal{D}_{\text{obs}}, \mathcal{D}_{\text{int}})$. Each $\mathcal{D}_{\text{int}}^{(i)}$ contains samples from the post-
 92 intervention distribution induced by replacing the mechanism of X_i with an exogenous P_{int} (a perfect
 93 stochastic intervention [13]). We write (N, M) for the balanced budget with N observational and M
 94 interventional samples per target, where $M = 0$ denotes no interventional data.

95 **Method classes.** A HYBRID-CD instance offers two admissible evidence channels: S and
 96 $(\mathcal{D}_{\text{obs}}, \mathcal{D}_{\text{int}})$. Methods are classified by which they consume. *Data-only* methods use only
 97 $(\mathcal{D}_{\text{obs}}, \mathcal{D}_{\text{int}})$ and include classical algorithms such as PC [34], GES [10], and ENCO [20]. *Semantic-*
 98 *only* methods use only S and include zero-shot LLM prompting from variable names. *Hybrid* methods
 99 use both, and most recent LLM-assisted pipelines fall in this class: they combine semantic priors
 100 with statistical summaries, candidate graphs, or edge constraints. Classical causal discovery and
 101 prompt-based LLM-CD are therefore limit cases of HYBRID-CD rather than separate settings, and
 102 whether a hybrid method actually benefits from the channels it receives is an empirical attribution
 103 question.

104 **Identifiability and references.** We adopt standard causal-discovery assumptions: causal sufficiency,
 105 no selection bias, acyclicity, Markov, and faithfulness [27]. These define what numerical evidence
 106 can identify in principle—observational data identify only the Markov equivalence class of G , while
 107 perfect stochastic interventions can break equivalences and, with sufficient target coverage, identify
 108 the full DAG in the limit [13]. The semantic channel S is not assumed correct; it may help, mislead,
 109 or carry no signal. Answering the attribution question on a fixed graph therefore requires three
 110 matched references: a semantic-only control at $N = M = 0$, a data-only classical baseline at the
 111 same (N, M) , and a semantic-cue control replacing S with anonymized placeholders. CAUSALMIX
 112 (Section 3) makes these references routine outputs of a single benchmark specification.

113 **Related benchmarks.** Existing benchmarks cover endpoints of HYBRID-CD but not the matched
 114 attribution problem. Causal-reasoning benchmarks test textual or symbolic causal reasoning [15, 43,
 115 8]; classical benchmarks test statistical recovery from numerical data [23, 9]; and LLM causal-graph
 116 benchmarks test graph construction from names or metadata [3, 32]. None of these holds a graph/data

condition fixed while jointly varying semantic cues, observational and interventional evidence, data representation, and matched semantic-only, hybrid, and data-only baselines—the joint control that motivates CAUSALMIX. Table 1 in Appendix C summarizes benchmark support for each control.

3 CAUSALMIX: Simulation and Evaluation Suite

CAUSALMIX operationalizes HYBRID-CD by turning graphs, data budgets, data representations, models, and baselines into matched evaluation cells. We write CAUSALMIX for the general suite and CAUSALMIX v1 for the fixed released instance used in this paper.

Design principles. CAUSALMIX is built around three principles that make attribution part of every measurement. (i) *Information sources are isolable*: semantic cues, data budget, and data representation are independently controlled axes, so recovery scores can be attributed to semantic priors, numerical evidence, representation effects, or their combination. (ii) *Cells are matched attribution tests*: within each comparison the graph and scoring code are held fixed and only the controlled axis varies, yielding semantic-only, hybrid, anonymized, and data-only cells under identical scoring. (iii) *Progress is measured against bracketing references*: a hybrid result is informative only when reported alongside the matched semantic-only floor, the anonymized control, and the matched data-only reference at the same data budget.

Cells and cell families. A benchmark specification contains graphs, semantic cue conditions, data budgets (N, M) , data representations, LLMs, data-only baselines, and seeds; a *benchmark cell* is one fixed combination of these fields plus a prompt template. Matched cells share all fields except the axis being compared, so one specification expands into four cell families: *semantic-only* (names, no data), *hybrid* (real names + data), *anonymized* (placeholder names + data), and *data-only* (classical methods on the same data). All cells are scored against the same ground-truth adjacency matrix with the same code, so every contrast changes exactly one information source at a time.

Simulation engine. For each graph G and budget (N, M) , the engine samples N observational examples from the parameterized joint distribution of G , and when $M > 0$ samples M interventional examples per target by perfect stochastic intervention [13, 20]: the target’s incoming edges are removed, the target is redrawn from a replacement distribution P_{int} , and downstream variables are resampled ancestrally. In CAUSALMIX v1’s bnlearn suite, $P_{\text{int}}(X_i) = \text{Uniform}(\mathcal{S}_i)$ over the state space of X_i ; this is configurable. Specifications that supply pre-sampled arrays (as in the Synthetic-25 tier) are treated as fixed data artifacts. Every generated cell records its source graph, data path, prompt, budget, and seed for exact regeneration.

Prompt conditions and data representations. The protocol separates semantic cues from data representations. Three semantic-cue conditions are defined: *names_only* (variable names only, no data), *real-name* (real names + data), and *anonymized* (placeholder names + the same data). Data-bearing cells use two representations: *summary*, encoding each observational or interventional regime as a sparse empirical histogram, and *tabular*, printing raw rows. Together, these axes define the matched contrasts that measure semantic dependence, data use, and representation sensitivity. Sections G.3 and H give templates and a token-scaling analysis.

Released instance. CAUSALMIX v1 locks graph sources, seeds, generated data, prompt templates, conditions, budgets, reference outputs, and scoring scripts. The release contains 14 graph instances in three tiers: six real reference graphs from bnlearn [33], six Synthetic-25 topology controls from ENCO [20], and two large challenge graphs. Section G.1 reports the full inventory, graph scale, and prompt-token burden.

4 Evaluation Setup

We evaluate CAUSALMIX v1 on Cancer, Earthquake, Asia, and Sachs using ten LLMs from the GPT-5, Llama-3.1, Qwen2.5, and Qwen3 families [25, 24, 12, 29, 30]. Each cell fixes the graph, semantic cue condition, data budget (N, M) , data representation, and seed; matched cells vary one axis. *names_only* is the semantic-only reference (names, no data); *real-name* and *anonymized* prompts hold data fixed and toggle whether variable names are real or placeholders, isolating name-mediated support. Classical references run on the same data without semantics: PC [34] and GES [10] are observational baselines ($M = 0$), and ENCO [20] is the interventional baseline. We also audit

168 five benchmark-adapted LLM-CD strategies—Causal-LLM and its data-driven branch [32], pairwise
169 prompting [38, 19], BFS-style querying [16], and SCP [37]—under the same controls.

170 **Metrics and progress bar.** We report directed-edge F1 (higher is better) and SHD (lower is better),
171 averaged over five independently sampled data realizations for data-bearing cells. A result constitutes
172 genuine progress on HYBRID-CD only if it improves over `names_only`, holds up under anonymiza-
173 tion, and moves toward the matched classical reference at the same graph and data budget. Full
174 protocol, response-processing rules, and model details are in Appendix D–F.

175 5 Benchmarking LLM-Assisted Causal Discovery

176 We use CAUSALMIX V1 to benchmark current LLMs on Cancer, Earthquake, Asia, and Sachs
177 and to evaluate a preliminary post-training procedure. Sachs is the largest of the four and supports
178 the denser budget sweeps and adapted-method audits. Unless otherwise noted, reported values are
179 directed-edge F1 with SHD in parentheses, averaged over valid parsed graph outputs; we refer to
180 $(N, M) = (5000, 200)$ as the primary budget.

181 The evaluation answers four research questions:

- 182 • **RQ1.** How much graph structure can LLMs recover from variable names alone, and does this
183 signal persist at larger graph scales?
- 184 • **RQ2.** When numerical evidence is supplied, are gains attributable to data rather than to names or
185 representation?
- 186 • **RQ3.** Does post-training with verifier-generated data improve evidence use under the same
187 controls?
- 188 • **RQ4.** How far are existing LLM-CD strategies from semantic-only and matched data-only
189 references?

190 RQ1, RQ2, and RQ4 demonstrate the diagnostic value of the suite (our first contribution); RQ3
191 reports preliminary results on the post-training procedure (our second contribution). All averages
192 exclude invalid parsed outputs; Table 4 reports the ten-model parse-validity profile, and the main
193 discussion emphasizes the GPT and Qwen2.5 slices with high parse validity.

194 5.1 RQ1: Recovery from Variable Names Alone

195 Under `names_only`, models see variable names but no samples. Figure 2 (left) shows that this signal
196 is already strong on small named graphs: across the 15 displayed cells for Cancer, Earthquake, and
197 Asia, seven are exact recoveries and ten have $F1 \geq 0.75$. Sachs is harder—all five displayed models
198 have $F1 \leq 0.51$ —but still yields nontrivial recovery. Larger models within a family typically recover
199 more from names alone: Qwen2.5-72B/14B/7B average 0.70/0.56/0.42 mean F1 across the four
200 graphs, and Llama-3.1-70B-Instruct averages 0.79 versus 0.38 for Llama-3.1-8B on Cancer and
201 Earthquake. This behavior is expected for familiar causal variables, since LLMs can infer plausible
202 relations from pretraining knowledge [28, 31, 6, 7]. Figure 2 (right) extends the probe to larger
203 graphs: GPT-5.2-Pro reaches $F1 = 0.78 \pm 0.03$ on Child (20 nodes) and 0.58 ± 0.05 on Alarm (37
204 nodes), while GPT-5-Mini reaches 0.51 ± 0.03 on Child and produces no valid Alarm graph. These
205 results show that raw LLM-CD scores on named public benchmarks can mix data-driven recovery
206 with semantic recall, and they motivate `names_only` as a routine control.

207 5.2 RQ2: Data Gains Depend on Semantic Cues and Representation

208 At the primary Sachs budget, adding numerical evidence does not produce a stable improvement
209 over the semantic-only baseline. Instead, the measured gain depends on how the same evidence is
210 presented and whether names are real or anonymized. Figure 3 shows observational and interventional
211 prompts can move F1 in either direction relative to `names_only`, and switching between summary and
212 tabular formats flips the result between paired cells. In the displayed cells, observational contrasts
213 span -0.04 to $+0.23$ F1 and interventional contrasts span -0.11 to $+0.20$. Bootstrap intervals in
214 Tables 10 and 11 serve as uncertainty annotations rather than high-powered tests; for GPT-5.2-Pro,
215 the real-summary prompt improves over names-only and exceeds its anonymized counterpart, while
216 the summary-tabular gap crosses zero.

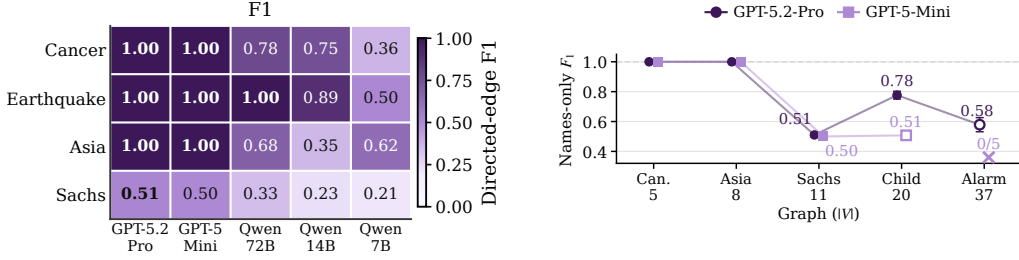


Figure 2: Names-only prompts recover substantial graph structure. Left: directed-edge F1 for names_only on the four main graphs; displayed models produced valid output for every cell, sorted by mean F1. Right: GPT-5.2-Pro and GPT-5-Mini names_only F1 on larger graphs, with standard errors over valid runs. Filled markers indicate all runs valid, hollow indicates partial validity, and × indicates no valid output.

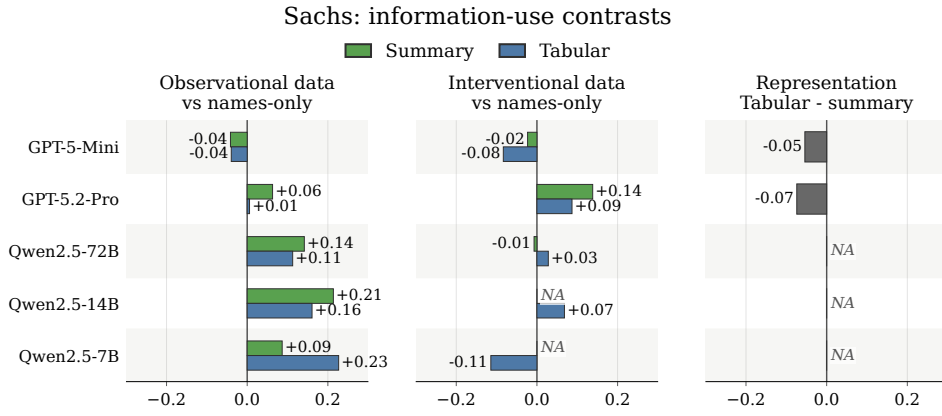


Figure 3: Matched information contrasts on Sachs. Bars show paired directed-edge F1 differences for adding samples relative to names_only and changing data format. Positive and negative contrasts show that measured data use depends jointly on the model and the prompt representation.

217 The anonymization test (Figure 10) further separates data use from name-mediated support: replacing
 218 real names with generic $X_1, \dots, X_d$ reduces F1 in 28 of 36 valid paired cells. Six of the eight
 219 non-negative cases occur on Sachs, where names-only recovery is already weak—when the model
 220 cannot infer much from names, dropping them costs little. Together, these results rule out a simple
 221 reading of raw LLM-CD F1: any data-grounded claim needs the semantic-only, anonymized, and
 222 format-matched controls that CAUSALMIX provides.

223 5.3 RQ3: Post-Training as a Step Toward Combining Priors and Data

224 The controls above identify a concrete repair target for open-weight LLM-CD systems: the model
 225 must emit a parseable graph, avoid relying on familiar names, and respond appropriately to numerical
 226 evidence. As a preliminary step beyond zero-shot prompting, we use CAUSALMIX’s own simulation
 227 engine to construct a verifier-based fine-tuning curriculum and post-train Qwen3-4B on it.

228 **Verifier curriculum.** The same generator that produces evaluation cells is reused as a training
 229 data factory. For each curriculum graph, we draw multiple (N, M) budgets and seeds, render
 230 names_only, summary, and tabular prompts under both real and anonymized names, and sample
 231 candidate completions from a base policy. A *ground-truth verifier* parses each completion’s adjacency
 232 matrix and scores it against the true graph by directed-edge F1; completions whose predicted graph is
 233 parseable, acyclic, and within an F1 tolerance of the verifier target are retained as supervised examples,
 234 while invalid or low-F1 completions are discarded. Because the verifier sees both observational
 235 and interventional prompts in the same data budget, the supervised set covers the full attribution
 236 grid (semantic-only, real-name data, anonymized data) rather than only one channel. We then

Effect of SFT + GRPO on Qwen3-4B-Thinking

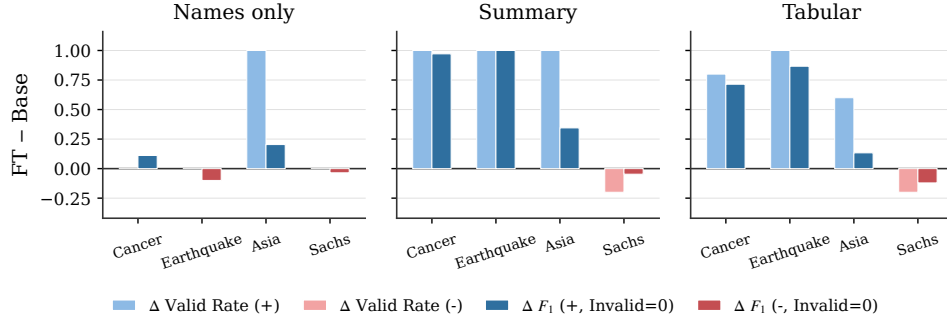


Figure 4: Effect of Qwen3-4B post-training. Bars report Qwen3-4B-FT minus the base Qwen3-4B model across graph and prompt conditions. Δ valid rate is the change in parseable-output rate; ΔF_1 is the change in directed-edge F1 with invalid outputs scored as zero.

fine-tune Qwen3-4B on the retained completions with standard supervised next-token training; full hyperparameters and curriculum statistics are in Appendix E.4.

Findings. Figure 4 reports the change in parse-valid response rate and directed-edge F1 (with invalid responses scored as zero) on the four evaluation graphs across all three prompt conditions. The fine-tuned model improves parse validity in nearly every cell and lifts directed-edge F1 on several names-only and data-bearing cells over the same base model, providing early evidence that targeted training can push LLMs toward combining semantic priors with numerical evidence rather than echoing semantic recall alone. Two caveats apply. First, the curriculum is not graph-disjoint from the four evaluation graphs, so improvements may partly reflect learning the output format and verifier preferences in addition to data-driven causal reasoning. Second, gains are concentrated on the smaller named graphs, where semantic priors are already largely sufficient; on Sachs the gains are smaller and a gap to ENCO remains. We therefore view this as preliminary evidence of promise—supportive of our second contribution—rather than a closed gap, and we leave broader curricula and held-out generalization to future work enabled by the released suite.

5.4 RQ4: Gap to Semantic and Data-Only References

We next ask whether existing LLM-based strategies improve beyond the semantic-only reference under a common protocol, by evaluating five LLM-CD strategies—Causal-LLM and its data-driven branch [32], pairwise prompting [38, 19], BFS-style querying [16], and SCP [37]—under our shared graph, data, and scoring protocol. Figure 5 shows that four of the five strategies fail to exceed names_only; SCP is the only exception, reaching $F_1 = 0.62$ (slightly above PC, below GES). Causal-LLM prompting reaches $F_1 = 0.49$, pairwise prompting 0.45, BFS-style 0.24, and the Causal-LLM data-driven branch 0.22.

When data match their assumptions, classical methods remain the strongest references. Figure 6 fixes Sachs at $N = 1000$ and increases M : the GPT-5.2-Pro real-name summary prompt moves from $F_1 = 0.51$ at $M = 0$ only to 0.56 – 0.57 at $M \in \{50, 100, 200\}$, while ENCO reaches $F_1 = 0.97$ – 1.00 once interventional samples are available. The relative strengths still depend on regime (Figure 7): GES benefits more than GPT-5.2-Pro from observational sample size, while GPT-5.2-Pro gains more than ENCO from intervention-only data—suggesting a complementary role for LLMs when a classical solver is outside its intended regime. CAUSALMIX makes this target measurable: future LLM-CD methods should improve beyond names_only, remain robust under anonymization, and move toward the matched data-only reference under the same graph and data conditions.

6 Discussion

The controls in CAUSALMIX recast LLM-CD evaluation as an attribution problem: a high score is informative only if it is paired with the names_only, anonymized, and matched data-only references that identify which information source produced it. One immediate implication is that high F1 on

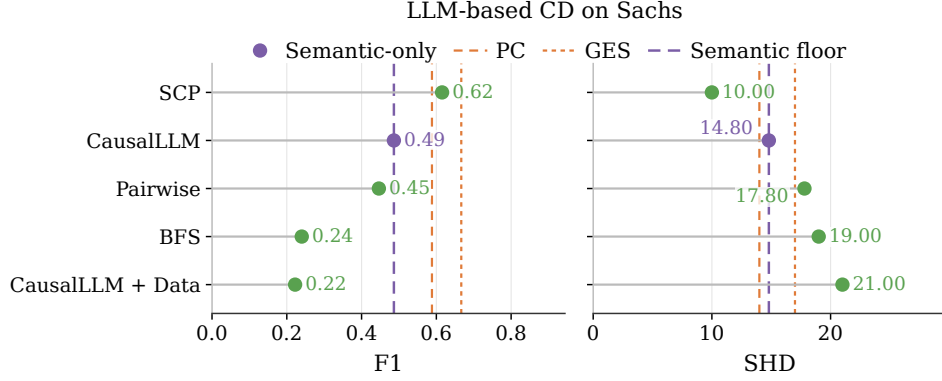


Figure 5: Adapted LLM-CD strategies on Sachs at $N = 5000$ observational samples [32, 38, 19, 16, 37], compared with names_only, PC, and GES under the same protocol.

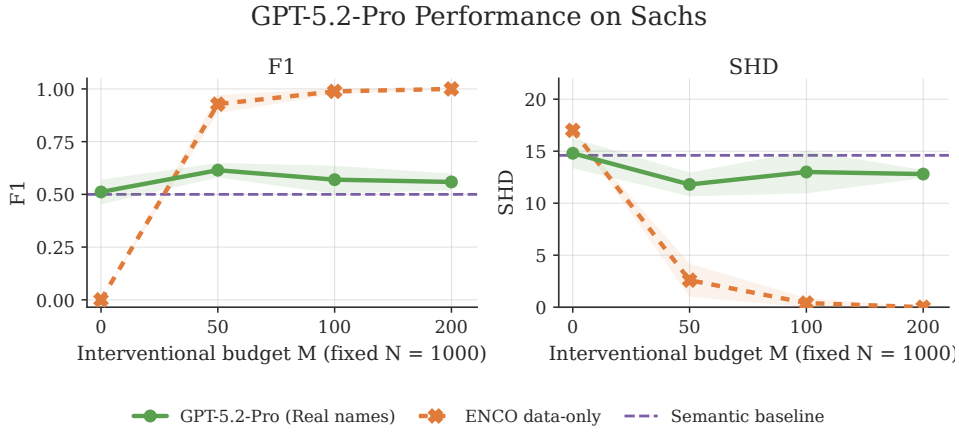


Figure 6: Sachs interventional-budget sweep with $N = 1000$ fixed. Increasing M has little effect on the GPT-5.2-Pro real-name summary prompt, while ENCO reaches near-perfect recovery once interventional samples are available.

named public benchmarks should not, by itself, be read as evidence of data-grounded causal discovery; long-used graphs that are easy to recover from semantics serve as stress tests for that distinction. The post-training experiment in Section 5.3 shows the promise and difficulty of verifier-generated training data: parse validity and recovery improve in several cells, while robust evidence use on harder graphs remains open. We see four research directions that the released suite is designed to support.

Combining evidence channels. Future LLM-CD systems need principled ways to combine statistical outputs, semantic judgments, and expert constraints. Classical methods can supply candidate graphs, separating sets, intervention effects, and uncertainty estimates; LLMs can translate domain knowledge into solver constraints, critique candidate edges, and request follow-up measurements. The open problem is aggregation *with attribution*: a hybrid system should report when expert constraints override statistical evidence, when classical outputs serve as priors, and whether iterative refinement improves reliability under matched controls.

Eliciting causal inductive biases. Names-only recovery and anonymization drops show that LLMs already bring strong priors. The useful question is how to convert those priors into testable causal hypotheses—acyclicity, intervention semantics, conditioning, confounding checks, sparsity—rather than unsupported edges. Algorithms can enforce these biases through edge queries, parent search, solver priors, or graph verifiers, and evaluation should test whether the elicited behavior remains stable under row, column, and edge-list permutations while still responding to real changes in sample size and intervention coverage.

GPT-5.2-Pro Performance on Sachs

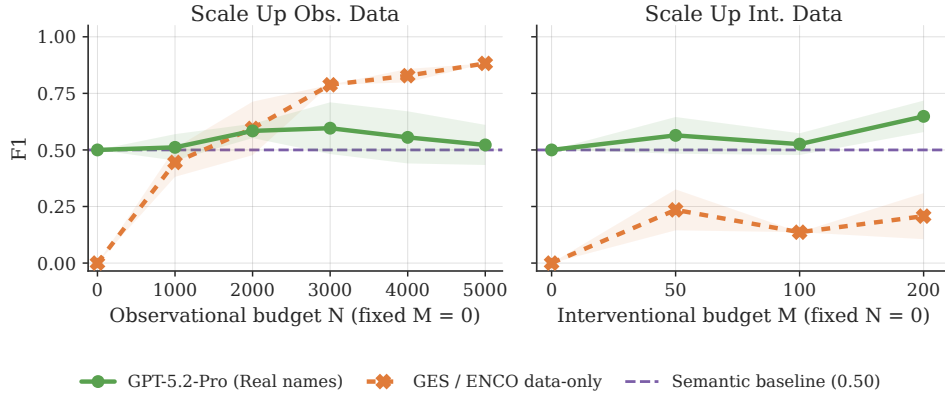


Figure 7: Observational-only and interventional-only sweeps on Sachs. Left ($M = 0$): GES improves with observational sample size, while the GPT-5.2-Pro summary prompt stays near names_only. Right ($N = 0$): interventional samples lift GPT-5.2-Pro above names_only, while ENCO is weak without observational samples.

291 **Compressing data into prompts.** Format sensitivity and context limits make data presentation a
 292 central problem. Raw tabular prompts scale with variables, samples, and intervention regimes; in our
 293 suite the 37-node Alarm graph reaches 1.4M tokens in the largest data-bearing representation, beyond
 294 most current context windows (Table 15). Useful representations may include empirical summaries,
 295 retrieval over variable blocks, local subgraph queries, uncertainty-ranked edge lists, and statistical
 296 front ends that translate raw samples into causal summaries. The observed summary–tabular gaps may
 297 reflect competing mechanisms—compression versus sample-level alignment—which the released
 298 format axis is designed to diagnose rather than to settle in favor of one serialization.

299 **Training with verifiable causal rewards.** Our verifier-based curriculum is one design point in a
 300 broader space. The key open question is what a causal-discovery verifier should reward. Single
 301 metrics such as F1, SHD, or acyclicity capture only part of the target; so do robustness checks for
 302 anonymization, intervention sensitivity, and representation stability. A stronger approach is multi-
 303 objective and comparative: train verifiers to choose which of two candidate graphs better explains
 304 the supplied evidence, respects expert constraints, and avoids unsupported edges, with curricula that
 305 progress across graph size, data budget, naming condition, and intervention coverage.

306 **Scope.** Our empirical claims are based on the four-graph slice (Cancer, Earthquake, Asia, Sachs); the
 307 smaller graphs have small graph spaces, so near-exact names_only recovery on them is a stress test
 308 rather than a capability claim. Larger CAUSALMIX V1 graphs (Child, Alarm, Synthetic-25, Diabetes,
 309 Pigs) are released as benchmark assets and scale tests but are not run on the full data-bearing grid
 310 here. Anonymized prompts probe name-mediated support but do not isolate broad domain knowledge
 311 from direct memorization; held-out and newly constructed named graphs are needed to push the
 312 diagnosis further. We treat these scope points as invitations to extend the suite rather than as closed
 313 limits of the framework.

314 7 Conclusion

315 We formalized hybrid causal discovery and released CAUSALMIX, a simulation and evaluation suite
 316 that turns the regime into matched attribution tests by varying semantic cues, data budgets, and data
 317 representations on a fixed graph alongside semantic-only and matched data-only references. Across
 318 Cancer, Earthquake, Asia, and Sachs, the suite shows that current LLMs recover substantial structure
 319 from variable names alone, that added data yields brittle and representation-sensitive gains, and that
 320 classical methods remain the strongest reference when the tabular problem is well specified. As a
 321 preliminary step beyond zero-shot prompting, a verifier-based post-training procedure on Qwen3-4B
 322 improves parse validity and recovery on several graphs, offering early evidence that targeted training
 323 can begin to combine semantic priors with numerical evidence—though a gap to classical methods
 324 remains. We hope CAUSALMIX becomes a shared standard for measuring whether future LLM-CD
 325 methods add useful, attributable, and transferable signal under matched evidence controls.

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A Broader Impacts

CAUSALMIX is an evaluation benchmark rather than a deployed decision-making system, so its expected impact is mainly on research practice. For causal discovery, the benchmark provides a way to distinguish performance that comes from variable-name semantics or domain priors from performance that comes from statistical evidence in the data. By reporting semantic-only controls, anonymized data-bearing settings, and matched data-only references, it can support more cautious interpretation of LLM-based causal discovery results and encourage future methods to demonstrate data-dependent recovery rather than only plausible graph generation.

For LLM research, the benchmark offers a concrete setting for studying how language models combine textual semantics, memorized domain knowledge, prompt formatting, and numerical data. The resulting comparisons can help identify when an apparent reasoning gain is better explained by shortcut use, sensitivity to data representation, or prior knowledge embedded in variable names. We therefore view the downstream contribution as methodological: improving the evaluation of LLMs on causal discovery tasks and providing diagnostics that may inform the design of future models, prompts, and benchmarks. These results should not be read as evidence that LLM-generated graphs are ready for use as causal conclusions without additional domain-specific validation.

B Limitations

Empirical scope. We run experiments on Cancer, Earthquake, Asia, and Sachs. The main attribution claims are based on this four-graph evaluated slice. Due to computational and prompt-token costs, we do not run all datasets in CAUSALMIX v1 across the complete matched grid. The remaining graphs, including Child, Alarm, Synthetic-25, and the large challenge graphs, are released as benchmark assets and scale tests; their graph scale and prompt-token burden are summarized in Table 15. We include a semantic-only Child/Alarm scale probe in the main text, but we do not yet run the full matched data-bearing grid on Child, Alarm, or Synthetic-25. The Sachs budget sweeps and method audits should therefore be treated as Sachs-specific scaling diagnostics, while the cross-graph attribution patterns should be treated as evidence about the four evaluated graphs rather than as a universal statement about all causal graphs, densities, or intervention designs.

Small-graph identifiability. Cancer and Earthquake have five nodes and four directed edges each, so their graph spaces are small and their named variables are highly diagnostic. Exact or near-exact names_only recovery on these graphs should not be read as evidence of general LLM causal-discovery ability. For observational data alone, directed DAG recovery is also limited by Markov equivalence unless additional assumptions or interventions orient the graph. This is why the evaluation reports semantic-only, anonymized, and classical references instead of treating raw F1 on the small graphs as a standalone capability claim.

Model and method coverage. The evaluated roster spans OpenAI, Llama, and Qwen families, but the strongest graph-recovery claims rely on the high-validity slices described in Section 4. Other closed-source models and future LLMs may interact differently with semantic controls, anonymization, and data formats. The audited LLM-CD variants are benchmark-adapted versions of representative query patterns, so the results test those patterns under matched conditions more than every engineering detail of the original systems. For hosted closed-source models, API model aliases may change over time; the released manifests record the model strings and access dates used for the reported cells, but future reruns may not be bitwise comparable.

Interpreting semantic controls. The names_only and anonymized conditions quantify dependence on semantic labels, but they do not identify the source of that dependence. High names-only recovery may reflect broad domain knowledge, benchmark memorization, or plausible but incorrect semantic priors. Anonymization also changes the naturalness of the prompt by replacing real variable names with placeholders. These controls are therefore attribution probes for name-mediated support, not definitive tests of training-data contamination. The main anonymization condition assigns placeholders in the declared variable order; future stress tests should randomize placeholder assignments and include held-out synthetic or newly constructed named graphs to better separate semantic priors from direct benchmark memorization.

Validity, prompting, and sample variation. Main graph-recovery averages exclude invalid parsed outputs and report validity separately; the post-training analysis additionally reports F1 with invalid outputs counted as zero. Both views are informative, but neither captures every way an interactive system might repair invalid outputs. We use deterministic decoding when available and five independently sampled data realizations for repeated data-bearing cells. Bootstrap intervals over the five runs are reported for representative Sachs matched contrasts, but this sample size is not enough to establish small effects of only a few F1 points with high power. The evaluation varies data format, naming condition, and data budget, but it does not exhaust minor prompt wording changes, variable-order perturbations, row-order perturbations, or repeated stochastic decodes.

Causal assumptions and external evidence. The analyzed benchmark cells assume causal sufficiency, a DAG, and generated observational/interventional samples from known graph param-

526 terizations. Latent confounding, cycles, selection bias, measurement error, and imperfect or soft
 527 interventions are outside the fixed analysis.

528 C Additional Related Work

529 Prior works on using LLMs for causal discovery generally fall into one of four categories:

- 530 1. **Direct inference:** the LLM outputs a causal graph (or causal order) as the primary estimate,
 531 typically from node metadata, natural-language descriptions, or summary information, with
 532 limited reliance on a separate discovery algorithm.
- 533 2. **LLMs as prior knowledge:** the LLM produces structured constraints or preferences (e.g.,
 534 required/forbidden edges, ordering constraints, edge priors) that are incorporated into a
 535 statistical causal discovery (SCD) procedure.
- 536 3. **Posterior refinement:** the LLM critiques, validates, or edits candidate graphs produced by
 537 traditional discovery methods as a post-processing step.
- 538 4. **Iterative graph refinement:** the LLM proposes graph revisions repeatedly in a search loop,
 539 where each revision is guided by feedback from an external evaluator computed from data.

540 Recent surveys summarize the rapid growth of this area from both the causal-discovery and broader
 541 causality-for-LLMs perspectives [39, 40, 18]. Our focus here is narrower: which existing *benchmarks*
 542 *and datasets* are closest to our setting, and why they still leave a gap for a controlled benchmark that
 543 evaluates how language models recover a causal graph from data.

544 C.1 Benchmark Comparison Summary

545 We compare benchmarks from causal reasoning, classical causal discovery, and LLM graph prediction.
 546 Causal-reasoning benchmarks test textual or symbolic causal reasoning; classical benchmarks test
 547 statistical recovery from numerical data; and LLM causal-graph benchmarks test graph construction
 548 from names, metadata, or limited summaries. CausalGraphBench [3] standardizes LLM graph
 549 construction from names and metadata, Causal-LLM [32] includes prompt-based and data-driven
 550 graph-discovery pipelines, and single-cell perturbation benchmarks [9] provide rich interventional
 551 measurements. These resources are complementary to Hybrid-CD evaluation, but none hold a single
 552 graph/data condition fixed while jointly varying semantic cues, data availability, data format, and
 553 semantic-only, hybrid, and data-only baselines. Table 1 summarizes this gap, which motivates
 554 CAUSALMIX.

Table 1: Benchmark controls for Hybrid-CD evaluation. ✓denotes substantial support for a control, and ✗denotes absence. Columns indicate whether a benchmark supports full graph discovery, semantic cues or controls, observational data, interventional data, multiple data formats, and matched semantic-only, hybrid, and data-only baselines.

Benchmark / Dataset	Graph CD	Semantic Ctrl.	Obs. Data	Int. Data	Data Format Ctrl.	Matched Baselines
CLadder [15]	✗	✓	✗	✗	✗	✗
Tübingen Cause–Effect Pairs [23]	✗	✗	✓	✗	✗	✗
CauSciBench [2]	✗	✓	✓	✗	✗	✗
CausalGraphBench [3]	✓	✓	✗	✗	✗	✗
Single-cell CausalBench [9]	✓	✗	✓	✓	✗	✗
LLM CausalBench [43]	✓	✓	✓	✗	✓	✗
Causal-LLM suite [32]	✓	✓	✓	✗	✓	✗
CAUSALMIX	✓	✓	✓	✓	✓	✓

555 C.2 Why Existing Benchmarks Are Insufficient

556 **Causal reasoning benchmarks for LLMs.** Recent benchmarks study whether LLMs can answer
 557 causal questions in text, math, or code settings. For example, Jin et al. [15] introduce *CLadder*, which
 558 textualizes symbolic causal queries and evaluates formal causal inference in natural language. Zhou
 559 et al. [43] introduce an LLM-oriented *CausalBench* to evaluate causal learning capabilities across
 560 correlation, skeleton recovery, and causality identification tasks. Chen et al. [8] introduce *CLEAR*,
 561 which studies language-model understanding of causal graphs through a structured benchmark

spanning multiple causal-graph tasks and complexity levels. At a broader level, Yang et al. [42] argue that many existing causal benchmarks for LLMs are vulnerable to shortcutting through background-knowledge retrieval rather than genuine causal reasoning, and recommend benchmarks with stronger construct validity grounded in intervention and counterfactual structure. These efforts are valuable, but they primarily evaluate *textual causal reasoning*. They do not evaluate the setting we target here: reconstructing an explicit causal graph from standardized observational and interventional data under controlled data formats and matched evaluation protocols.

Benchmarking LLM causal graph discovery. The closest prior benchmark to our setting is *CausalGraphBench* [3]. *CausalGraphBench* assembles 35 causal graphs from public repositories and papers, augments them with graph-level metadata such as domain descriptions and clarified node dictionaries, and compares several LLM-based graph-construction strategies under a unified evaluation. This is an important step toward standardizing evaluation for LLM causal graph discovery. However, its task setup remains primarily *metadata-driven*: methods operate from node names and benchmark metadata rather than from matched observational and interventional data conditions. In addition, the benchmark is released as a fixed collection rather than as a reusable framework for defining new benchmark settings. Our work is complementary but different in emphasis: we target controlled benchmarks in which models infer a causal graph from data, explicit controls over name-based shortcuts (real names, anonymized names, and names-only cues), matched classical baselines, and benchmark specifications that can generate benchmark families over both curated real graphs and large controlled synthetic graph sets.

Roy et al. [32] present a unified one-shot framework for prompt-based and data-driven causal graph discovery, and their released codebase includes synthetic DAG/data generation and evaluation components. Their contribution is therefore closer to a method-and-pipeline framework than to a benchmark whose primary purpose is controlled attribution of information sources. Our focus is different: HYBRID-CD formalizes causal discovery when semantic side information, observational data, and optional interventional data may all be informative, and CAUSALMIX is designed to hold the causal task fixed while varying data budget, data format, and semantic cue condition. This makes it possible to compare semantic-only, data-only, and hybrid methods under matched graph and data conditions, and to ask not only whether a method recovers the graph, but which information source is responsible for that performance.

LLMs as assistants for causal discovery. A second line of work uses LLMs as auxiliary components inside causal discovery pipelines rather than as the main subject of a benchmark. Long et al. [21] introduced early pairwise LLM-CD by querying GPT-3 to score candidate directed edges between variable pairs in medical DAGs. Long et al. [22] later use language models as imperfect experts to orient edges within a Markov equivalence class. Abdulaal et al. [1] and Takayama et al. [37] combine metadata-driven LLM judgments with data-driven fitting or statistical discovery in iterative refinement loops. More recently, Dong et al. [11] show that pretrained LLMs can over-rely on variable-name semantics and introduce a supervised fine-tuning setup to better integrate statistical summaries with language priors. These papers demonstrate that LLMs may help as semantic priors or expert-like assistants, but they do not provide a reusable benchmark for *end-to-end LLM causal discovery from shared data conditions*. In particular, they do not isolate how data format, semantic cue condition, and names-only semantic baselines affect graph recovery within one matched evaluation setup.

Classical causal discovery benchmarks. On the classical side, benchmark design has advanced substantially, especially in biology. Mooij et al. [23] provide the Tübingen cause–effect pairs, which are seminal for bivariate causal direction inference. However, Tübingen evaluates pairwise direction rather than full-DAG recovery, so it is best viewed as background classical prior art rather than as a direct benchmark comparator for our setting. Chevalley et al. [9] introduce a large-scale benchmark for network inference from single-cell perturbation data, motivated in part by the insufficiency of synthetic-only evaluation for real systems. However, that benchmark targets large-scale biological network inference rather than LLM-based graph prediction. It is not designed around data formats for language models, and because the true graph is not directly available, it relies on biological and distributional proxy metrics rather than exact recovery against a shared DAG target. As a result, it provides important realism, but it does not answer the evaluation question we study.

C.3 Direct Inference

Roy et al. [32] propose *Causal-LLM*, a pipeline that generates the causal graph by one of two variants: a *prompting* variant that gives an LLM a structured description of the dataset (node/variable metadata) and asks it to list directed edges, and a tabular data variant that passes samples through a frozen LLM backbone with learned projection heads to score directed edges. In both cases, they apply post-processing to prune weak edges and enforce acyclicity. Experiments are reported on synthetic and real benchmarks at multiple graph sizes (up to 100 nodes).

Early pairwise LLM-CD was introduced by Long et al. [21], who queried GPT-3 to score candidate directed edges between variable pairs in medical DAGs. Kıcıman et al. [19] subsequently benchmarked LLMs on pairwise causal discovery tasks, and Susanti and Färber [35] includes knowledge graphs in the prompt for pairwise causal discovery.

Most similar to our setup, Susanti and Färber [36] provides observational data in prompts but uses either pairwise or BFS queries to construct the full graph. Its BFS variant is closely related to Jiralerspong et al. [16]: both perform breadth-first graph construction with LLM queries, but differ in how observational evidence is represented in the prompt.

Other works in this category include Vashishtha et al. [38], Jiralerspong et al. [16].

C.4 LLMs as Prior Knowledge

Ban et al. [5] uses LLMs as structured prior knowledge to guide score-based causal structure learning, while explicitly addressing the fact that LLM causal reasoning can be noisy. Their pipeline first asks the LLM to derive richer variable metadata, then extracts a small set of high-confidence causal statements, maps these to ancestral path constraints, and uses them as soft constraints for score-based structure learning.

C.5 Posterior Refinement

Long et al. [22] use LLMs as imperfect experts to shrink the Markov equivalence class by providing edge-direction judgments from metadata.

Dong et al. [11] observe that pretrained LLMs often default to inferring edges from variable-name semantics while underusing observational evidence, producing graphs that are unreliable under perturbations like variable-name permutation or column reordering. The authors introduce CARE, a supervised fine-tuning paradigm that trains an LLM to integrate algorithmic causal discovery outputs with world knowledge to produce more robust, data-grounded causal graphs.

Khatibi et al. [17] propose ALCM, which first generates an initial graph from observational data using standard discovery algorithms (PC, LiNGAM, or NOTEARS), then converts candidate edges into causal-aware prompts that inject contextual information and dataset metadata, and uses an LLM-driven refiner to validate edges, revise orientations, remove spurious links, and add missing links.

C.6 Iterative Graph Refinement

LLM-first. Abdulaal et al. [1] propose *Causal Modelling Agent* (CMA), which alternates between an LLM and a model-based fit criterion. CMA starts from an empty graph; the LLM proposes a candidate causal graph from dataset metadata, which is compiled into a computational graph and scored by fitting a deep structural causal model (DSCM) to the data. The LLM subsequently edits the graph in two stages—global structural changes, then individual edge revisions—after which the DSCM is refit, and the loop repeats.

Model-first. Takayama et al. [37] bootstrap from a conventional statistical causal discovery estimate. They first run SCD to obtain an initial graph, then prompt an LLM with these statistical outputs (*statistical causal prompting*, SCP) to elicit pairwise causal judgments with calibrated probabilities. These probabilities are converted into structured constraints and fed back into SCD to produce an updated graph, repeating the SCD→LLM→constraints→SCD loop.

Peng et al. [26] propose *CauScientist*, which uses a hybrid initialization by selecting between an LLM-generated graph and a conventional SCD baseline according to an intervention-aware BIC

664 criterion. Starting from the selected graph G_0 , the method runs a propose–verify–update loop: at step
 665 t , the LLM proposes a single atomic edit (add/delete/reverse); an external verifier checks structural
 666 validity and BIC improvement; if both checks pass the candidate is accepted, otherwise the state is
 667 unchanged.

668 Other works in this category include Ban et al. [4].

669 D Evaluation Protocol and Metrics

670 D.1 Reliability Metrics

671 For LLM-based causal discovery, a benchmark cell is not fully characterized by a single decoded
 672 graph because data-bearing prompts are instantiated on multiple independently sampled datasets.
 673 For each condition cell, the framework therefore records replicate summaries in addition to mean
 674 graph-recovery metrics. These include the acyclic-output fraction, dispersion of per-run F1 and SHD,
 675 edge-stability summaries across repeated predictions, and a consensus graph formed from empirical
 676 edge frequencies at a fixed threshold. Consensus graphs are scored with the same graph-recovery
 677 metrics as individual responses, yielding consensus F1, consensus SHD, consensus skeleton and
 678 ancestor scores, and consensus acyclicity. When probability-like edge frequencies are available, the
 679 framework also reports directed and skeleton Brier scores.

680 D.2 Reporting and Aggregation

681 All LLMs are prompted to return a JSON object containing the predicted adjacency matrix. The
 682 output parser extracts the $d \times d$ integer matrix, verifies that all entries are binary, and flags outputs
 683 that are not parseable. Primary tables report condition-level averages over valid parsed responses;
 684 format compliance is tracked separately so that instruction-following failures do not masquerade as
 685 causal-discovery errors. Invalid responses include outputs that omit a graph, use variables outside
 686 the declared variable set, return non-binary matrices, or fail to follow the requested JSON format.
 687 They are excluded from graph-recovery averages and counted as parse-validity failures. Main figures
 688 report error bars over valid parsed runs only and visually mark conditions with fewer than two valid
 689 parsed runs, since such cells measure format reliability as much as graph-recovery quality. For
 690 matched contrasts with reported confidence intervals, we compute the contrast at the seed level
 691 whenever the same data-realization index is present in both cells, then draw percentile bootstrap
 692 intervals over those paired differences. Only pairs for which both cells produce a valid parsed
 693 graph and a nonempty directed-edge F1 score are used. The Sachs uncertainty tables are generated
 694 by `scripts/build_sachs_bootstrap_tables.py`, which reads the canonical-condition manifest,
 695 loads the corresponding per-row metric sidecars, and runs percentile bootstrap resampling with a
 696 fixed seed and 20,000 resamples. Because the main repeated cells use five data realizations, these
 697 intervals are best read as uncertainty annotations that prevent over-interpreting small differences, not
 698 as a substitute for a high-powered evaluation.

699 For matched-cell contrasts, the framework first aggregates repeated runs within each condition cell
 700 and then compares cells that differ along one controlled axis. Thus hybrid gain, anonymization
 701 drop, data-format sensitivity, observational-data uplift, and intervention uplift are computed from
 702 condition-level summaries rather than from unmatched individual responses. When a metric is
 703 undefined for a graph or condition, such as v-structure F1 on a ground-truth graph with no unshielded
 704 colliders, the corresponding cell is left undefined rather than forced to zero. Predicted graphs are
 705 scored as emitted: no post-hoc cycle removal or other graph repair is applied before evaluation.

706 D.3 Metric Reference

707 **Contrastive metrics.** Let $S_{g,\theta}(c, f, N, M)$ denote a condition-level score, such as mean directed-
 708 edge F1, for graph g , model θ , semantic cue condition $c \in \{\text{real}, \text{anon}\}$, data representation
 709 $f \in \{\text{summary}, \text{tabular}\}$, observational budget N , and intervention budget M per target. Let
 710 $S_{g,\theta}^{\text{name}}$ denote the corresponding semantic-only names_only score. All contrasts below hold g and θ
 711 fixed and are computed only when both matched cells are available.

712 • **Mixed-information gain** compares a data-bearing prompt with the semantic-only reference:

$$\Delta_{\text{mix}}(c, f, N, M) = S_{g,\theta}(c, f, N, M) - S_{g,\theta}^{\text{name}}.$$

Table 2: Unit metrics in CAUSALMIX. These metrics score one predicted graph or repeated runs within one benchmark cell.

Metric	Interpretation
Edge F1	Higher values indicate better directed-edge recovery.
SHD	Lower values indicate fewer edge additions, deletions, or reversals needed to match the ground-truth DAG.
Acyclicity	Higher values indicate that raw model outputs already satisfy the DAG constraint.
Skeleton F1	Higher values indicate better recovery of adjacent variable pairs, ignoring orientation.
Ancestor F1	Higher values indicate better recovery of causal reachability relations.
Run mean / variance	Mean summarizes typical performance; variance summarizes stochastic reliability within one benchmark cell.

713 Positive values indicate improvement beyond names alone.

714 • **Anonymization drop** compares real-name and anonymized prompts at the same data representa-
715 tion and budget:

$$\Delta_{\text{name}}(f, N, M) = S_{g,\theta}(\text{real}, f, N, M) - S_{g,\theta}(\text{anon}, f, N, M).$$

716 Positive values indicate a real-name advantage.

717 • **Data-format gap** compares summary and tabular prompts at the same semantic cue condition
718 and data budget:

$$\Delta_{\text{fmt}}(c, N, M) = S_{g,\theta}(c, \text{summary}, N, M) - S_{g,\theta}(c, \text{tabular}, N, M).$$

719 Larger magnitudes indicate stronger sensitivity to prompt representation.

720 • **Observational uplift** compares a positive observational budget $n > 0$ with the no-observational-
721 data cell while holding M , c , and f fixed:

$$\Delta_{\text{obs}}(c, f; n | M) = S_{g,\theta}(c, f, n, M) - S_{g,\theta}(c, f, 0, M).$$

722 • **Intervention uplift** compares a positive intervention budget $m > 0$ with the no-intervention cell
723 while holding N , c , and f fixed:

$$\Delta_{\text{int}}(c, f; m | N) = S_{g,\theta}(c, f, N, m) - S_{g,\theta}(c, f, N, 0).$$

724 E Additional Empirical Results

725 This section collects the full cross-graph result views that support the compressed figures and claims in
726 the main text. Values are directed-edge F1 unless otherwise stated; missing cells indicate unavailable
727 runs, invalid matched cells, or prompts that exceeded the context window. All experiments are done
728 on 2 H200 GPUs.

729 E.1 Cross-Graph Coverage

730 **Prediction validity.** Table 4 reports the parse-valid response ratio at the primary budget after select-
731 ing the highest-validity prompt configuration tested for each graph, model, and data representation.

732 **Classical baseline grid.** Tables 6 to 9 report all available PC, GES, and ENCO cells for the four
733 evaluated graphs and all evaluated data-budget combinations in the source baseline artifact.

Table 3: Coverage of the four-graph LLM sweep. Both configuration files instantiate the same matched information cells and differ only in prompt wrapper and format-hint policy.

Axis	Coverage	#
<i>Evaluation subjects</i>		
Graphs	Cancer, Earthquake, Asia, Sachs	4
Models	GPT-5-Mini, GPT-5.2-Pro; Qwen2.5-7B/14B/72B; Qwen3-4B/30B-A3B; Llama-3.1-8B/8B-Instruct/70B-Instruct	10
<i>Protocol axes</i>		
Prompt-wrapper suites	chat + format hint; plain, no hint	2
Matched conditions per suite	names-only $\cup$ {real, anon} $\times$ {summary, tabular} $\times$ budgets	17
Data budgets (N, M)	(5000, 200), (5000, 0), (0, 200), (1000, 0)	4
Naming regimes	real names; anonymized	2
Data representations	summary; tabular / matrix serialization	2
Total graph $\times$ model $\times$ suite $\times$ condition cells		4 $\times$ 10 $\times$ 2 $\times$ 17 = 1,360

Table 4: Valid response ratio after selecting the highest-validity prompt configuration tested at $N = 5000$, $M = 200$. Each cell reports Names/Sum/Tab, where entries are valid/total runs. “–” denotes prompts that exceed the context window.

Model	Cancer			Earthquake			Asia			Sachs		
	Names	Sum	Tab	Names	Sum	Tab	Names	Sum	Tab	Names	Sum	Tab
GPT-5-Mini	5/5	5/5	5/5	5/5	5/5	5/5	5/5	3/5	0/5	5/5	5/5	5/5
GPT-5.2-Pro	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5
Qwen3-4B	5/5	0/5	0/5	5/5	0/5	0/5	0/5	0/5	0/5	5/5	5/5	5/5
Qwen3-30B-A3B	5/5	0/5	0/5	5/5	0/5	1/5	0/5	0/5	0/5	5/5	2/5	1/5
Qwen2.5-7B	5/5	4/5	5/5	5/5	5/5	4/5	5/5	5/5	2/5	5/5	0/5	3/5
Qwen2.5-14B	5/5	5/5	5/5	5/5	5/5	4/5	5/5	5/5	5/5	5/5	5/5	1/5
Qwen2.5-72B	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	5/5	–
Llama-3.1-8B	5/5	4/5	5/5	5/5	2/5	5/5	0/5	4/5	5/5	0/5	1/5	–
Llama-3.1-8B-Instruct	5/5	5/5	4/5	0/5	5/5	4/5	0/5	5/5	0/5	0/5	1/5	–
Llama-3.1-70B-Instruct	5/5	5/5	5/5	5/5	5/5	5/5	0/5	5/5	0/5	0/5	0/5	–

Table 5: Available cross-graph matched controls at the reported budget. Values are directed-edge F1. Rows hold graph and model fixed while varying semantic cue and data format; the final column reports the strongest matched data-only reference when available.

Graph	Model	Names	Real summ.	Anon summ.	Real tabular	Anon tabular	Data-only
Cancer	GPT-5-Mini	0.92	1.00	0.51	1.00	0.70	ENCO 0.86
Earthquake	GPT-5-Mini	1.00	1.00	0.91	1.00	0.86	–
Asia	GPT-5-Mini	1.00	1.00	0.58	–	–	ENCO 0.93
Sachs	GPT-5-Mini	0.50	0.45	0.44	0.37	0.32	ENCO 1.00

Table 6: All available classical baseline cells for Asia. Values are directed-edge F1 and SHD for PC, GES, and ENCO at each evaluated observational/interventional budget. A dash indicates that F1 is undefined in the source metric file, typically because the method returned no positive directed edges; SHD is still reported.

Graph	Method	N	M	F1	SHD
Asia	ENCO	0	200	0.23	16.80
Asia	ENCO	200	3	0.50	10.00
Asia	ENCO	1000	0	–	8.00
Asia	GES	1000	0	0.77	3.40
Asia	PC	1000	0	0.38	8.20
Asia	ENCO	1000	50	0.83	3.00
Asia	ENCO	1000	100	0.86	2.40
Asia	ENCO	1000	200	0.91	1.40
Asia	ENCO	5000	0	–	8.00
Asia	GES	5000	0	0.76	3.80
Asia	PC	5000	0	0.70	4.20
Asia	ENCO	5000	200	0.96	0.60
Asia	ENCO	50000	512	1.00	0.00

Table 7: All available classical baseline cells for Cancer. Values are directed-edge F1 and SHD for PC, GES, and ENCO at each evaluated observational/interventional budget.

Graph	Method	N	M	F1	SHD
Cancer	ENCO	0	3	0.18	9.00
Cancer	ENCO	0	50	0.29	5.00
Cancer	ENCO	0	100	0.25	6.00
Cancer	ENCO	0	200	0.24	7.60
Cancer	ENCO	200	0	–	4.00
Cancer	ENCO	200	3	–	7.00
Cancer	ENCO	200	5	–	7.00
Cancer	ENCO	500	5	0.40	3.00
Cancer	ENCO	1000	0	–	4.00
Cancer	GES	1000	0	0.29	5.20
Cancer	PC	1000	0	0.33	4.60
Cancer	ENCO	1000	50	0.56	3.20
Cancer	ENCO	1000	100	0.64	2.40
Cancer	ENCO	1000	200	0.77	2.20
Cancer	ENCO	2000	30	0.57	3.00
Cancer	ENCO	5000	0	–	4.00
Cancer	GES	5000	0	0.71	2.20
Cancer	PC	5000	0	0.43	4.20
Cancer	ENCO	5000	200	0.91	0.60
Cancer	ENCO	50000	512	1.00	0.00
Cancer	ENCO	50000	2000	0.86	1.00

Table 8: All available classical baseline cells for Earthquake. Values are directed-edge F1 and SHD for PC, GES, and ENCO at each evaluated observational/interventional budget.

Graph	Method	N	M	F1	SHD
Earthquake	ENCO	0	200	0.32	6.80
Earthquake	ENCO	1000	0	–	4.00
Earthquake	GES	1000	0	0.95	0.40
Earthquake	PC	1000	0	0.57	4.60
Earthquake	ENCO	1000	50	0.89	1.00
Earthquake	ENCO	1000	100	0.98	0.20
Earthquake	ENCO	1000	200	0.98	0.20
Earthquake	ENCO	5000	0	–	4.00
Earthquake	GES	5000	0	1.00	0.00
Earthquake	PC	5000	0	1.00	0.00
Earthquake	ENCO	5000	200	1.00	0.00

Table 9: All available classical baseline cells for Sachs. Values are directed-edge F1 and SHD for PC, GES, and ENCO at each evaluated observational/interventional budget.

Graph	Method	N	M	F1	SHD
Sachs	ENCO	0	50	0.24	30.80
Sachs	ENCO	0	100	0.14	38.00
Sachs	ENCO	0	200	0.21	32.40
Sachs	ENCO	0	500	0.13	40.00
Sachs	ENCO	1	0	–	17.00
Sachs	ENCO	1	50	0.11	17.00
Sachs	ENCO	1	100	0.11	16.00
Sachs	ENCO	1	200	0.11	17.00
Sachs	ENCO	100	0	–	17.00
Sachs	GES	100	0	0.38	15.60
Sachs	PC	100	0	0.33	14.20
Sachs	ENCO	100	10	0.55	13.00
Sachs	ENCO	100	50	0.67	9.00
Sachs	ENCO	100	100	0.64	9.00
Sachs	ENCO	100	200	0.69	8.00
Sachs	ENCO	100	500	0.58	10.00
Sachs	GES	500	0	0.45	15.40
Sachs	PC	500	0	0.34	17.40
Sachs	ENCO	500	20	0.74	9.00
Sachs	ENCO	1000	0	–	17.00
Sachs	GES	1000	0	0.44	17.00
Sachs	PC	1000	0	0.42	16.80
Sachs	ENCO	1000	50	0.93	2.60
Sachs	ENCO	1000	100	0.99	0.40
Sachs	ENCO	1000	200	1.00	0.00
Sachs	ENCO	1000	500	0.97	1.00
Sachs	GES	2000	0	0.60	13.00
Sachs	PC	2000	0	0.50	16.00
Sachs	GES	3000	0	0.79	7.00
Sachs	PC	3000	0	0.48	17.60
Sachs	GES	4000	0	0.83	5.80
Sachs	PC	4000	0	0.60	13.60
Sachs	ENCO	5000	0	–	17.00
Sachs	GES	5000	0	0.88	4.00
Sachs	PC	5000	0	0.60	13.60
Sachs	ENCO	5000	50	0.95	1.80
Sachs	ENCO	5000	100	0.97	1.00
Sachs	ENCO	5000	200	1.00	0.00
Sachs	ENCO	5000	500	1.00	0.00
Sachs	ENCO	8000	0	–	17.00
Sachs	ENCO	8000	50	0.97	1.00
Sachs	ENCO	8000	100	0.97	1.00
Sachs	ENCO	8000	200	1.00	0.00
Sachs	ENCO	8000	500	1.00	0.00

734 **E.2 Evidence Ladders and Matched Contrasts**

735 **Bootstrap uncertainty.** Tables 10 and 11 report percentile bootstrap intervals for matched Sachs
736 contrasts at the primary budget.

737 **E.3 Sachs Numeric Sweep Tables**

738 The main text uses figures to emphasize the Sachs scaling trends. The following tables give the
739 corresponding numeric values for the observational and interventional sweeps.



Figure 8: Full semantic-only heatmap across the ten-model roster. Cells report directed-edge F1 for names_only prompting, with no observational or interventional data. Gray cells marked 0/5 indicate that none of the five runs produced a valid parsed graph output.

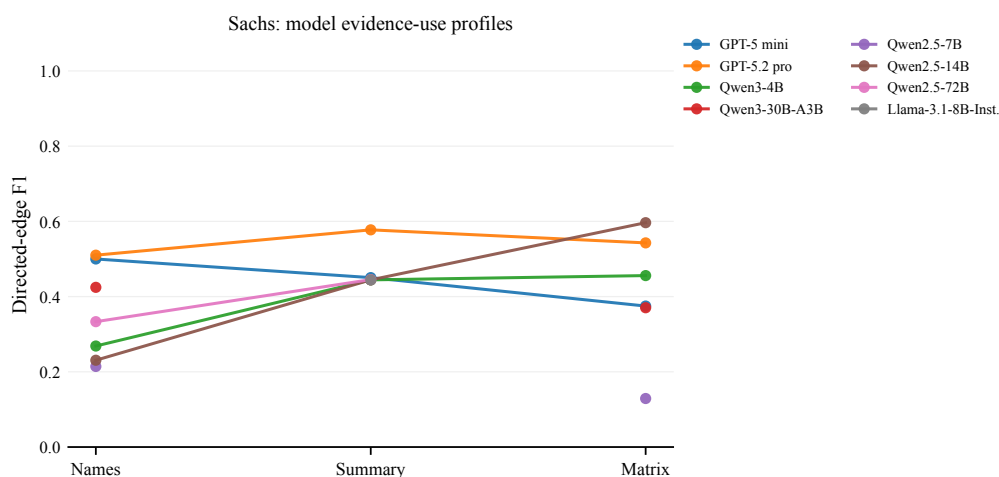


Figure 9: Cross-graph matched-control profile for one representative model at the reported budget. Each line holds the graph fixed and compares semantic-only prompting, data-bearing prompting, and the matched classical reference where available.

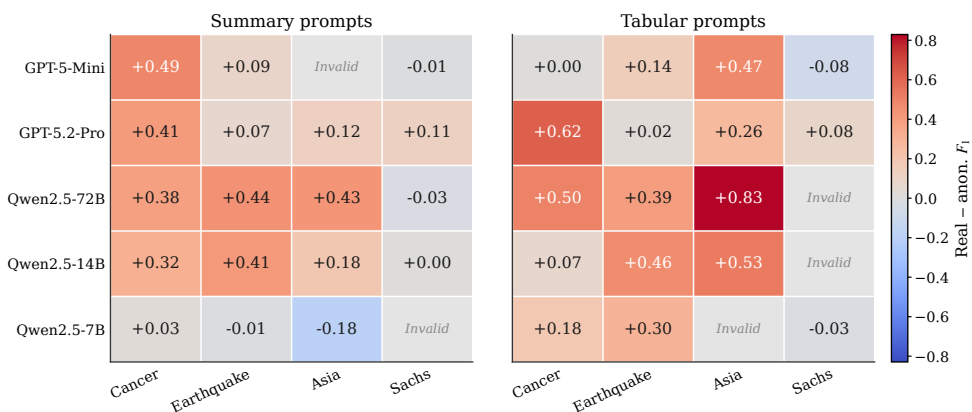


Figure 10: Effect of replacing real variable names with anonymized names. Each entry shows anonymized-name F1 minus real-name F1 for a paired evaluation cell. Negative values mean real names helped; values near zero mean little dependence on names; positive values mean the anonymized prompt performed better. Gray entries lack valid outputs for one or both paired settings.

Sachs

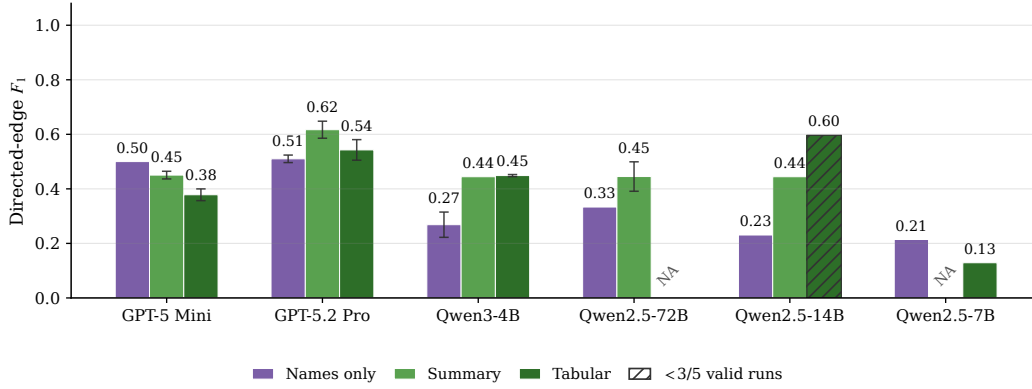


Figure 11: Full evidence ladder for pretrained models on Cancer, Earthquake, Asia, and Sachs with real variable names. Each panel holds the graph fixed and compares the semantic-only reference with the available data-bearing prompt conditions and matched data-only references.

Cancer

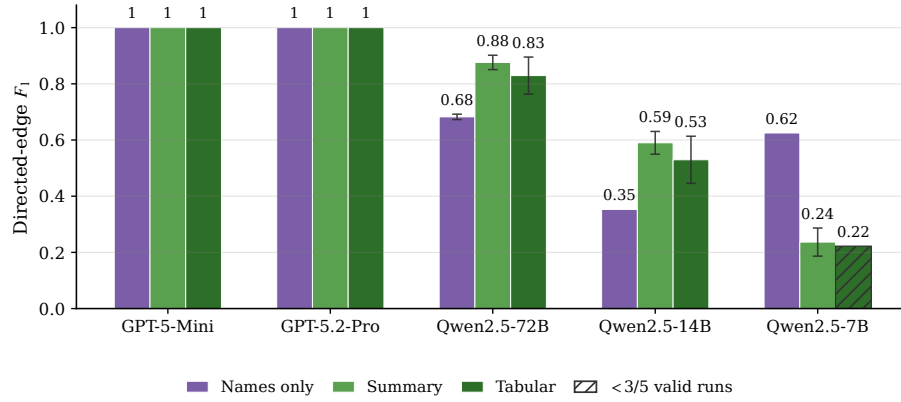


Earthquake



Figure 12: Per-graph evidence ladders for pretrained models on Cancer and Earthquake with real variable names. These panels expand the corresponding rows in Figure 11.

Asia



Sachs



Figure 13: Per-graph evidence ladders for pretrained models on Asia and Sachs with real variable names. These panels expand the corresponding rows in Figure 11.

Sachs — anonymized

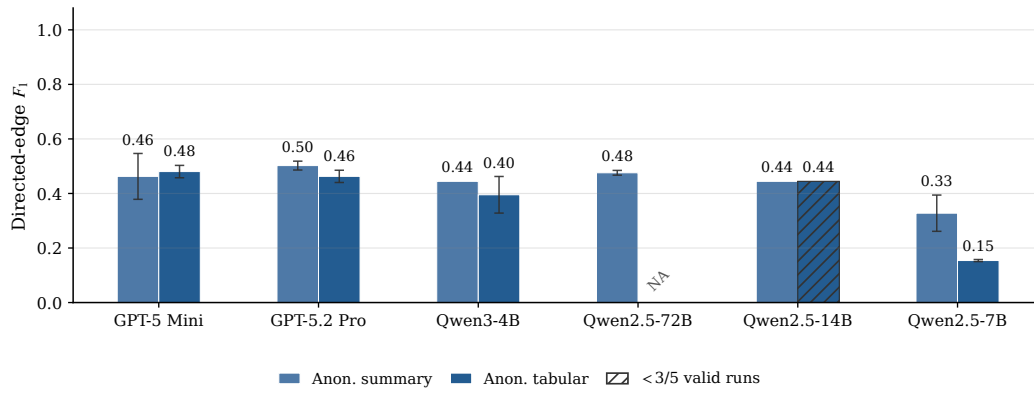


Figure 14: Full evidence ladder for pretrained models under anonymized variable names. Comparing this figure with Figure 11 shows which data-bearing cells remain strong when name semantics are removed.

Table 10: Bootstrap uncertainty for primary Sachs matched contrasts for GPT-5.2-Pro at $(N, M) = (5000, 200)$. Intervals are percentile bootstrap intervals over paired valid runs using 20,000 resamples. Contrasts pair runs by the same data-realization index; with five seeds, these intervals are descriptive uncertainty annotations rather than high-powered significance tests.

Value name	Value formula	Paired runs	Mean ΔF_1	95% bootstrap interval
Data gain	real summary – names-only	5	+0.107	[+0.057, +0.158]
Anon. drop	real summary – anon summary	5	+0.115	[+0.055, +0.167]
Format gap	real summary – real tabular	5	+0.074	[−0.014, +0.189]

Table 11: Bootstrap uncertainty for primary Sachs matched contrasts for the remaining nine evaluated models. The GPT-5.2-Pro companion table is shown in Table 10. Intervals use 20,000 percentile-bootstrap resamples over paired valid seed-level F1 differences. Cells with fewer than two paired valid rows are not estimable and are marked “–”.

Model	Value name	Value formula	Paired runs	Mean ΔF_1	95% bootstrap interval
GPT-5-Mini	Data gain	real summary – names-only	5	−0.050	[−0.074, −0.025]
	Anon. drop	real summary – anon summary	5	−0.012	[−0.148, +0.128]
	Format gap	real summary – real tabular	5	+0.054	[−0.022, +0.139]
Qwen3-4B	Data gain	real summary – names-only	5	+0.176	[+0.090, +0.249]
	Anon. drop	real summary – anon summary	4	+0.000	[+0.000, +0.000]
	Format gap	real summary – real tabular	4	−0.004	[−0.013, +0.000]
Qwen3-30B-A3B	Data gain	real summary – names-only	0	---	---
	Anon. drop	real summary – anon summary	0	---	---
	Format gap	real summary – real tabular	0	---	---
Qwen2.5-7B	Data gain	real summary – names-only	0	---	---
	Anon. drop	real summary – anon summary	0	---	---
	Format gap	real summary – real tabular	0	---	---
Qwen2.5-14B	Data gain	real summary – names-only	3	+0.214	[+0.214, +0.214]
	Anon. drop	real summary – anon summary	2	+0.000	[+0.000, +0.000]
	Format gap	real summary – real tabular	1	---	---
Qwen2.5-72B	Data gain	real summary – names-only	5	+0.112	[+0.021, +0.203]
	Anon. drop	real summary – anon summary	5	−0.031	[−0.121, +0.057]
	Format gap	real summary – real tabular	0	---	---
Llama-3.1-8B	Data gain	real summary – names-only	0	---	---
	Anon. drop	real summary – anon summary	0	---	---
	Format gap	real summary – real tabular	0	---	---
Llama-3.1-8B-Instruct	Data gain	real summary – names-only	0	---	---
	Anon. drop	real summary – anon summary	1	---	---
	Format gap	real summary – real tabular	0	---	---
Llama-3.1-70B-Instruct	Data gain	real summary – names-only	0	---	---
	Anon. drop	real summary – anon summary	0	---	---
	Format gap	real summary – real tabular	0	---	---

Table 12: Sachs observational-budget sweep for GPT-5-Mini at fixed $M = 0$. Values are F1 (SHD), with PC and GES as matched observational references.

N	Names	Real summ.	Anon summ.	PC	GES
0	0.50 (14.20)	–	–	–	–
1000	–	0.51 (14.60)	0.52 (13.20)	0.33 (20.00)	0.49 (21.00)
5000	–	0.47 (16.40)	0.62 (12.20)	0.59 (14.00)	0.67 (17.00)

Table 13: Sachs intervention-budget sweep for GPT-5.2-Pro at fixed $N = 1000$. Values are F1 (SHD); GES is used at $M = 0$ and ENCO when interventions are available.

M	Real summ.	Anon summ.	Data-only ref.	Data-only F1 (SHD)
0	0.51 (14.80)	0.47 (15.20)	GES	0.49 (21.00)
50	0.57 (13.20)	0.43 (15.40)	ENCO	1.00 (0.00)
100	0.57 (13.00)	0.48 (15.00)	ENCO	0.97 (1.00)
200	0.56 (12.80)	0.47 (14.60)	ENCO	0.97 (1.00)

E.4 Post-Training

For the RQ3 post-training experiment, we perform a two-stage supervised LoRA adaptation of Qwen/Qwen3-4B-Thinking-2507. Both stages use PEFT LoRA with rank 16, scaling factor $\alpha = 16$, dropout 0.0, no bias terms, and adapters applied to q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. Training uses standard next-token supervision. The verifier is used only to construct and validate supervised targets; it is not used as an RL reward for this checkpoint.

Stage 1: Format and graph-target SFT. We first train a causal-discovery format adapter, qwen3_4b_format. The training samples covers the small causal graphs cancer, earthquake, asia, and sachs, with observational/interventional budgets $(N, M) \in \{100, 1000\} \times \{10, 50\}$. Each completion is required to follow the structured contract `<think> ... </think><answer>{"adjacency_matrix": ...}</answer>`, where the answer matrix is the simulator’s ground-truth DAG. We train for 2 epochs on two GPUs using bfloat16, fused AdamW, learning rate 2×10^{-5} , no warmup, no weight decay, max gradient norm 1.0, per-device batch size 1, and gradient accumulation 4. Checkpoints are evaluated every 100 optimizer steps on a held-out split, and the checkpoint with the lowest validation loss is reloaded at the end of training. The goal of this step is to teach the model output format.

Stage 2: Reasoning SFT continuation. Starting from the Stage 1 adapter, we continue SFT on format_sft_5way_v4_small.jsonl. This dataset is built from sft_source_v4_small.csv, which contains 768 prompt-answer rows over earthquake, asia, and sachs; prompt styles summary and matrix; budgets $(N, M) \in \{100, 1000\} \times \{10, 50\}$; real and anonymized names; random row and column order; four column permutations; and prompt-level reasoning guidance set to either none or concise. Each source row is expanded into five supervision styles: no-guidance, answer-only, no-guidance concise evidence, no-guidance teacher evidence, concise-guidance concise evidence, and concise guidance for teachers. This yields 3840 total SFT records.

Teacher-evidence records use Qwen/Qwen2.5-72B-Instruct-AWQ as a local OpenAI-compatible teacher to draft only the rationale. The final `<answer>` payload remains the verifier’s gold adjacency matrix. Teacher rationales are retried or discarded if they fail format or edge-consistency validation.

For the exported root checkpoint, Stage 2 uses a 5% validation split (3648 training records and 192 validation records). We train on two GPUs using bfloat16, fused AdamW, learning rate 5×10^{-6} , linear decay with no warmup, no weight decay, max gradient norm 1.0, per-device batch size 1, and gradient accumulation 8. Checkpoints are saved and evaluated every 100 optimizer steps, and model selection uses the lowest validation loss. The root adapter corresponds to the validation-best export at step 200 (validation loss 0.39997); later diagnostic subcheckpoints are retained in the same directory but are not used for the reported RQ3 result.

F Model and Baseline Details

Model sweep. The evaluated model roster spans four model families: one hosted OpenAI family and three open-weight families. Within each open-weight family, we include multiple model sizes where available to support within-family comparisons under the same benchmark cells.

- **OpenAI:** GPT-5-Mini (gpt-5-mini) and GPT-5.2-Pro (gpt-5.2-pro) [25, 24]. The public model strings and per-call access dates are stored in the released manifests; the cited OpenAI system-card references were accessed on 2026-05-03.
- **Qwen3:** Qwen3-4B (Qwen/Qwen3-4B-Thinking-2507) [30], and Qwen3-30B-A3B (Qwen/Qwen3-30B-A3B-Thinking-2507) [30].
- **Qwen2.5:** Qwen2.5-7B (Qwen/Qwen2.5-7B-Instruct-1M) [41], Qwen2.5-14B (Qwen/Qwen2.5-14B-Instruct-1M) [41], and Qwen2.5-72B (Qwen/Qwen2.5-72B-Instruct-AWQ) [29].
- **Llama-3.1:** Llama-3.1-8B (meta-llama/Meta-Llama-3.1-8B) [12], Llama-3.1-8B-Instruct (meta-llama/Meta-Llama-3.1-8B-Instruct) [12], and Llama-3.1-70B-Instruct (meta-llama/Llama-3.1-70B-Instruct) [12].

Decoding and replication. Prompt-based LLM calls use temperature 0 whenever the inference endpoint supports deterministic decoding. If an endpoint does not expose temperature 0, we use its

Table 14: Information regimes for the main causal-discovery methods discussed in Section C.

Method	Semantic Info	Obs. Data	Int. Data
Data-only methods			
PC [34]	✗	✓	✗
GES [10]	✗	✓	✗
ENCO [20]	✗	✓	✓
Causal-LLM (data-driven) [32]	✗	✓	✗
Semantic-only methods			
names_only	✓	✗	✗
Causal-LLM (prompting) [32]	✓	✗	✗
Knowledge-graph structure prompting [35]	✓	✗	✗
Pairwise causal prompting [38, 19]	✓	✗	✗
Mixed-information methods			
Causal graph prompting with observational data [36]	✓	✓	✗
Efficient BFS-style querying [16]	✓	✓	✗
LLM-derived structured prior knowledge [5]	✓	✓	✗
MEC posterior refinement [22]	✓	✓	✗
Statistical causal prompting (SCP) [37]	✓	✓	✗
CauScientist [26]	✓	✓	✓

fixed supported decoding setting and keep that setting fixed across matched cells. For data-bearing prompts, the five replicates are independently sampled data realizations rather than repeated stochastic decodes of one prompt.

Information regimes. Table 14 organizes the main causal-discovery methods by the information channels they use. Data-only methods serve as matched reference bars, semantic-only methods define the semantic-only reference, and hybrid methods are LLM-method variants that combine semantic cues with supplied data.

Data-only references. PC [34] is a constraint-based method that removes edges using conditional-independence tests and then orients the remaining edges with graphical rules. GES [10] is a score-based greedy equivalence-search procedure over Markov equivalence classes. ENCO [20] parameterizes edge existence and orientation separately and uses observational plus interventional data to recover a fully directed graph beyond the observational MEC. PC and GES are applied to observational data; ENCO is applied to paired observational and interventional data. Classical methods are run once per (N, M) cell.

Benchmark-adapted LLM-method variants. The semantic Causal-LLM prompting baseline receives only variable names and the task instruction and predicts the full DAG in one shot. The BFS-style baseline receives observational summary prompts, asks for likely root variables, and iteratively queries for direct children to grow the graph frontier. SCP [37] uses the PC-based observational branch: it runs PC, converts LLM pairwise judgments into required or forbidden edge constraints, and reruns constrained PC. In our setup, BFS-style querying and SCP use semantic information plus observational evidence, but no interventional data.

G Dataset and Prompt Details

G.1 Released Dataset Details

CAUSALMIX V1 is the fixed release used for comparable reporting. It locks graph sources, random seeds, generated observational and interventional data, prompt templates, semantic cue conditions, data representations, data budgets, reference outputs, and scoring scripts. These locked fields make reported scores reproducible and make matched-cell contrasts comparable across semantic-only, anonymized, data-bearing, and data-only conditions.

Table 15: Graph scale and prompt-token burden in CAUSALMIX v1. Density is $|E|/[d(d-1)]$ for a directed graph with d nodes. Tok. mean and Tok. max summarize prompt-token counts for the largest data-bearing representation.

Dataset	Nodes	Edges	Density	Tok. mean	Tok. max
Real reference suite					
Cancer	5	4	0.200	14.1K	84.6K
Earthquake	5	4	0.200	13.8K	84.6K
Asia	8	8	0.143	25.4K	152.7K
Sachs	11	17	0.154	62.7K	232.2K
Child	20	25	0.066	193.0K	541.3K
Alarm	37	46	0.035	433.7K	1.4M
Synthetic topology suite					
Chain	25	24	0.040	137.1K	781.4K
Collider	25	24	0.040	137.1K	781.4K
Bidiag	25	47	0.078	137.1K	781.4K
Jungle	25	46	0.077	137.1K	781.4K
Random	25	96	0.160	137.1K	781.4K
Full	25	300	0.500	137.1K	781.4K
Large challenge suite					
Diabetes	413	602	0.004	60.2M	336.8M
Pigs	441	592	0.003	46.5M	126.1M

Table 16: Controlled axes in the generated Hybrid-CD dataset. Each benchmark cell fixes a graph and chooses one value from each axis, enabling matched contrasts between semantic priors, data use, and data-format sensitivity.

Axis	Values	What the contrast isolates
Semantic cue	real names; anonymized names	Whether performance depends on semantic variable names rather than supplied data.
Data budget	no data: (0, 0) observational only: (1000, 0), (5000, 0) interventional only: (0, 200) mixed: (1000, 50), (1000, 200), (5000, 50), (5000, 200)	Whether additional observational or interventional samples improve graph recovery.
Data format	summary, tabular	Whether the same causal information leads to different conclusions under different data presentations.
Graph family	real Bayesian networks; synthetic topology controls; large challenge graphs	Whether the benchmark behavior persists across named real systems, controlled topology stress tests, and scale limits.

819 The release contains 14 graph instances in three tiers: six real reference graphs from bnlearn [33], six
820 Synthetic-25 topology controls from ENCO exports [20], and two large challenge graphs. The main
821 paper evaluates four real graphs—Cancer, Earthquake, Asia, and Sachs—with Sachs supporting the
822 densest budget and method-audit grids. Diabetes and Pigs are included as graph- and prompt-scaling
823 stress tests, not as primary single-prompt LLM evaluation cells. The release includes generated
824 prompt CSVs, manifests, source-graph references, generated observational and interventional data,
825 prompt templates, scoring code, and matched PC, GES, and ENCO baseline outputs where applicable.
826 Main metrics are directed-edge F1, SHD, acyclicity, skeleton recovery, ancestor recovery, and
827 matched-cell contrast metrics. Future challenge splits can instantiate the same generator on held-out
828 graphs, seeds, or graph families while preserving the matched-cell protocol.

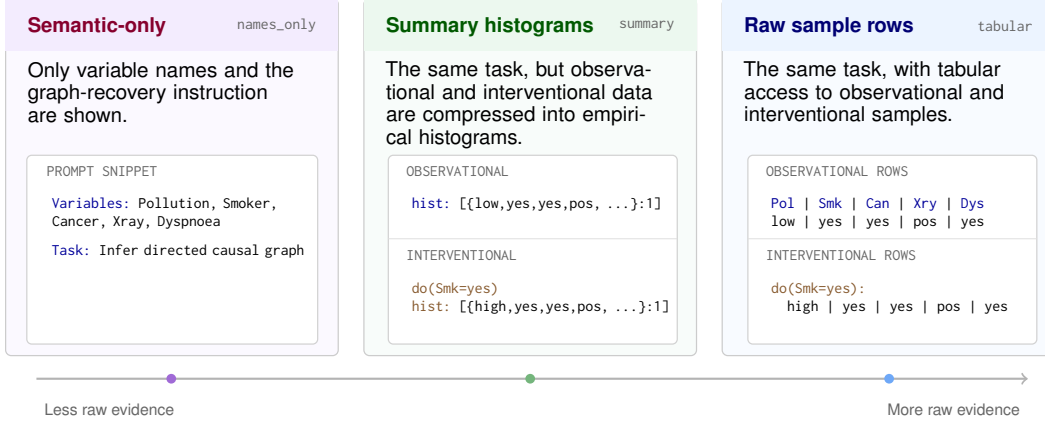


Figure 15: Prompt representations in CAUSALMIX. All three ask for the same graph-recovery output while varying the exposed evidence: variable names only, summary histograms, or raw sample rows. $\text{do}(X=v)$ marks post-intervention regimes generated by the stochastic intervention procedure; full prompts are in the appendix.

G.2 Controlled Dataset Axes

G.3 Prompt Conditions and Representations

The protocol separates *semantic cue conditions* from *data representations*. It generates three semantic cue conditions. In `names_only` cells, the model sees variable names and the task description, with no supplied data. In `real-name` cells, the model sees original variable names with supplied data. In `anonymized` cells, the model sees anonymized variable names with the same supplied data. For data-bearing cells, the protocol uses two representations: `summary`, which provides compressed observational and interventional empirical histograms, and `tabular`, which provides raw tabular rows. Figure 15 shows what each condition or representation exposes; the released templates provide the full prompt strings. Full templates and examples appear in Section H.

names_only. This no-data probe defines the semantic-only reference against which hybrid gains must be interpreted. The `names_only` prompt provides only the variable names and the task description, with no data. It is the semantic-only baseline.

summary. This representation tests whether a model can use compressed numerical data under a controlled token budget. The `summary` prompt represents each observational or interventional regime as a sparse empirical histogram over full variable assignments: each entry gives one observed assignment and its count. For interventions, the prompt maps labels such as $\text{do}(X=v)$ to the corresponding post-intervention histogram. This removes repeated rows while preserving the empirical regimes used by the benchmark. Section G.4 gives the token-scaling analysis.

tabular. This representation tests whether explicit tabular access exposes signals that histogram summaries may compress away. The `tabular` prompt presents data in a tabular representation in which rows correspond to samples and columns correspond to variables. Because it prints raw rows, it is useful for auditing representation sensitivity on smaller graphs and becomes a stress test as graph or sample size grows.

G.4 Prompt Serialization and Token Scaling

The `summary` prompt encodes each observational or interventional regime as a sparse empirical joint histogram over full variable assignments. Each histogram entry lists an assignment in the declared variable order and its count. For interventional data, the prompt maps each regime label, such as $\text{do}(X=v)$, to the corresponding post-intervention histogram.

Let d be the number of variables, k_j the number of states of variable j , $K = \prod_{j=1}^d k_j$ the number of possible full assignments, n_r the number of samples in regime r , and h_r the number of distinct as-

signments observed in regime r . The prompt size scales as $\mathcal{O}\left(d \sum_{r=1}^R h_r\right)$, where $h_r \leq \min(n_r, K)$ and R is the number of data regimes shown in the prompt: one observational regime plus the included interventional regimes. Thus summary removes repeated rows rather than printing every sample.

In the large-graph sparse regime with roughly k states per variable and $k^d \gg N, M$, almost every observed full assignment is unique, so $h_r \approx n_r$. The token burden is then dominated by the number of retained samples per regime and the length- d assignment vector printed for each distinct assignment, giving $\mathcal{O}(d \sum_r n_r)$ before truncation. With a fixed cap of H retained histogram entries per regime, or with frequency/mass truncation, the displayed size is bounded by $\mathcal{O}(dRH)$. For comparison, tabular prints raw rows and scales as $\mathcal{O}\left(d \sum_{r=1}^R n_r\right)$. This makes summary a compact audit format for categorical data while preserving the empirical regimes used by the benchmark.

870 G.5 Prompt Wrapper and Format Hint

The four-graph LLM sweep uses two prompt-wrapper suites with identical underlying task/data content. The plain suite sends the base prompt string directly and does not append an additional format hint. The chat suite appends the format hint below to the user prompt, wraps the result in the following transcript form, and pre-fills the assistant turn with `<think>`.

Chat wrapper. For a base prompt denoted by `<BASE_PROMPT>`, the exact serialized chat wrapper is:

```
system
A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant
→ first thinks about the solution and then provides the final answer.
user
<BASE_PROMPT>

Formatting requirement: Output exactly: <think>...</think><answer>{"adjacency_matrix":
→ [[0,1,...],[0,0,...],...]}</answer> where the matrix is N×N with integer entries 0 or 1 in VARIABLES order,
→ and [i][j]=1 means variable i directly causes variable j.
assistant
<think>
```

Format hint. The exact line appended to the user prompt when `append_format_hint` is enabled is:

```
Formatting requirement: Output exactly: <think>...</think><answer>{"adjacency_matrix":
→ [[0,1,...],[0,0,...],...]}</answer> where the matrix is N×N with integer entries 0 or 1 in VARIABLES order,
→ and [i][j]=1 means variable i directly causes variable j.
```

877 H Prompt Templates and Full Examples

To make the benchmark concrete, we include the literal prompt templates for the semantic-only condition and the two data-bearing formats. The examples use the Cancer graph with a small number of samples for readability; benchmark runs use the same templates with the data budgets specified in the released manifests. The main text figure abstracts these surface forms, while this appendix shows the underlying literal prompt strings. We also experimented with template-based tabular serialization inspired by TabLLM [14], in which each row is rendered as a natural-language template rather than as a compact tabular row. In our causal-discovery prompts, this serialization used more tokens than the tabular format below and degraded graph-recovery performance, so the released benchmark uses the more compact tabular representation for raw rows. A likely explanation is that template serialization allocates many tokens to repeated linguistic scaffolding rather than to sample values. For causal discovery, the useful signal lies in cross-row and cross-variable distributional patterns. The compact tabular format preserves row/column alignment and exposes more observations per token, whereas TabLLM-style natural-language templates break this alignment, reduce the amount of data that fits in context, and may increase reliance on semantic wording rather than empirical evidence.

892 H.1 names_only Template

Because `names_only` is the no-data semantic probe, its prompt is much shorter than the two data-bearing formats.

```
ROLE: You are an expert in causal discovery from variable semantics and background knowledge.
TASK: Infer the directed causal graph over the variables.
We are studying a system called '<DATASET_NAME>'.
```

No observational or interventional data are provided for this case. Use only the variable meanings and relevant
 $\hookrightarrow$ background causal knowledge.

ASSUMPTIONS:

- The true graph is a DAG (no directed cycles).
- Causal sufficiency holds (no unobserved confounders among these variables).

--- VARIABLE ORDER (ORDER MATTERS) ---

0: <V1>
1: <V2>
...
d-1: <Vd>

--- AVAILABLE EVIDENCE ---

No sampled data are provided. The graph must be inferred from the variable names and domain knowledge alone.

--- OUTPUT INSTRUCTIONS ---

Output exactly: <think>...</think><answer>...</answer>.
Keep <think> concise (minimal necessary reasoning only).
Inside <answer>, output exactly one JSON object with key "adjacency_matrix".
- "adjacency_matrix": N x N 0/1 matrix in declared variable order.
No extra text before, between, or after the two blocks.
The JSON in <answer> must start with "{" and end with "}".

895 H.2 names_only Example

ROLE: You are an expert in causal discovery from variable semantics and background knowledge.
TASK: Infer the directed causal graph over the variables.
We are studying a system called 'cancer'.
No observational or interventional data are provided for this case. Use only the variable meanings and relevant
 $\hookrightarrow$ background causal knowledge.

ASSUMPTIONS:

- The true graph is a DAG (no directed cycles).
- Causal sufficiency holds (no unobserved confounders among these variables).

--- VARIABLE ORDER (ORDER MATTERS) ---

0: Pollution
1: Smoker
2: Cancer
3: Xray
4: Dyspnoea

--- AVAILABLE EVIDENCE ---

No sampled data are provided. The graph must be inferred from the variable names and domain knowledge alone.

--- OUTPUT INSTRUCTIONS ---

Output exactly: <think>...</think><answer>...</answer>.
Keep <think> concise (minimal necessary reasoning only).
Inside <answer>, output exactly one JSON object with key "adjacency_matrix".
- "adjacency_matrix": N x N 0/1 matrix in declared variable order.
No extra text before, between, or after the two blocks.
The JSON in <answer> must start with "{" and end with "}".

896 H.3 summary Template

ROLE: You are an expert in causal discovery from observational and interventional data.
TASK: Infer the directed causal graph over the variables.
The following are empirical distributions computed from data sampled from a Bayesian network named <DATASET_NAME>.

ASSUMPTIONS:

- The true graph is a DAG (no directed cycles).
- Causal sufficiency holds (no unobserved confounders among these variables).
- Interventions are perfect do-interventions (surgical): $\text{do}(X=v)$ cuts all incoming edges into X.

--- VARIABLE ORDER (ORDER MATTERS) ---

0: <V1>
1: <V2>
...
d-1: <Vd>

--- DATA FORMAT ---

Each payload summarizes empirical samples in the declared variable order.
num_states=[<k1>, <k2>, ..., <kd>].
hist entries use [assignment, count, prob]; assignments follow the declared variable order.
observational_data summarizes samples with no intervention.
interventional_data maps each $\text{do}(X=v)$ regime to its empirical summary.
Unlisted assignments may be absent due to finite sampling.

--- OBSERVATIONAL DATA ---

observational_data={
 "n": <N_obs>,
 "hist": [
 [<x1>, <x2>, ..., <xd>], <count>, <prob>],

```

    ...
  ]
}

--- INTERVENTIONAL DATA ---
interventional_data={
  "do(<X>=<value>)": {
    "n": <N_int>,
    "hist": [
      [<x1>, <x2>, ..., <xd>], <count>, <prob>],
      ...
    ]
  },
  ...
}

--- END OF DATA ---

--- OUTPUT INSTRUCTIONS ---
Output exactly: <think>...</think><answer>...</answer>.
Keep <think> concise (minimal necessary reasoning only).
Inside <answer>, output exactly one JSON object with key "adjacency_matrix".
- "adjacency_matrix": N x N 0/1 matrix in declared variable order.
No extra text before, between, or after the two blocks.
The JSON in <answer> must start with "{" and end with "}".

```

897 H.4 summary Example

ROLE: You are an expert in causal discovery from observational and interventional data.
TASK: Infer the directed causal graph over the variables.
The following are empirical distributions computed from data sampled from a Bayesian network named cancer.
ASSUMPTIONS:
- The true graph is a DAG (no directed cycles).
- Causal sufficiency holds (no unobserved confounders among these variables).
- Interventions are perfect do-interventions (surgical): $\text{do}(X=v)$ cuts all incoming edges into X .

--- VARIABLE ORDER (ORDER MATTERS) ---
0: Pollution
1: Smoker
2: Cancer
3: Xray
4: Dyspnoea

--- DATA FORMAT ---
Each payload summarizes empirical samples in the declared variable order.
num_states=[2,2,2,2,2].
hist entries use [assignment, count, prob]; assignments follow the declared variable order.
observational_data summarizes samples with no intervention.
interventional_data maps each $\text{do}(X=v)$ regime to its empirical summary.
Unlisted assignments may be absent due to finite sampling.

--- STATE LEGEND ---
Pollution: 0->low, 1->high
Smoker: 0->False, 1->True
Cancer: 0->False, 1->True
Xray: 0->negative, 1->positive
Dyspnoea: 0->True, 1->False

--- OBSERVATIONAL DATA ---
observational_data={
 "n": 10,
 "hist": [[0,0,0,0,1],2,0.2],[0,0,0,1,1],2,0.2],[0,0,0,0,0],1,0.1],[0,0,0,1,0],1,0.1],[1,0,0,1,0],1,0.1],[
 ↪ [1,0,0,0,1],1,0.1],[0,1,0,1,1],1,0.1]]
}

--- INTERVENTIONAL DATA ---
interventional_data={
 "do(Cancer=False)": {"n":3,"hist":[[0,1,0,0,0],1,0.3333],[0,1,0,0,1],1,0.3333],[1,1,0,0,0],1,0.3333]],
 "do(Cancer=True)": {"n":3,"hist":[[0,0,1,1,1],1,0.3333],[0,1,1,1,0],1,0.3333],[0,0,1,1,0],1,0.3333]],
 "do(Dyspnoea=False)": {"n":3,"hist":[[0,0,0,0,1],2,0.6667],[0,1,0,0,1],1,0.3333]],
 "do(Dyspnoea=True)": {"n":3,"hist":[[0,1,0,1,0],1,0.3333],[1,1,0,0,0],1,0.3333],[0,0,0,0,0],1,0.3333]],
 "do(Pollution=high)": {"n":3,"hist":[[1,1,0,0,0],1,0.3333],[1,0,0,1,0],1,0.3333],[1,0,0,0,1],1,0.3333]],
 "do(Pollution=low)": {"n":3,"hist":[[0,0,0,0,1],1,0.3333],[0,1,0,0,0],1,0.3333],[0,0,0,1,1],1,0.3333]],
 "do(Smoker=False)": {"n":3,"hist":[[0,0,0,0,1],1,0.3333],[1,0,0,0,0],1,0.3333],[0,0,0,1,1],1,0.3333]],
 "do(Smoker=True)": {"n":3,"hist":[[0,1,0,0,1],2,0.6667],[0,1,0,0,0],1,0.3333]],
 "do(Xray=negative)": {"n":3,"hist":[[1,0,0,0,1],1,0.3333],[0,1,0,0,1],1,0.3333],[0,0,0,0,1],1,0.3333]],
 "do(Xray=positive)": {"n":3,"hist":[[0,0,0,1,1],1,0.3333],[0,1,0,1,1],2,0.6667]]
}

--- END OF DATA ---

--- OUTPUT INSTRUCTIONS ---

Output exactly: <think>...</think><answer>...</answer>.
 Keep <think> concise (minimal necessary reasoning only).
 Inside <answer>, output exactly one JSON object with key "adjacency_matrix".
 - "adjacency_matrix": N x N 0/1 matrix in declared variable order.
 No extra text before, between, or after the two blocks.
 The JSON in <answer> must start with "{" and end with "}".

898 H.5 tabular Template

ROLE: You are an expert in causal discovery from observational and interventional data.
 TASK: Infer the directed causal graph over the variables.
 The following are empirical samples from a Bayesian network named <DATASET_NAME>.
 ASSUMPTIONS:
 - The true graph is a DAG (no directed cycles).
 - Causal sufficiency holds (no unobserved confounders among these variables).
 - Interventions are perfect do-interventions (surgical): do(X=v) cuts all incoming edges into X.

--- VARIABLE ORDER (ORDER MATTERS) ---
 0: <V1>
 1: <V2>
 ...
 d-1: <Vd>

--- OBSERVATIONAL DATA (no intervention) ---
 Each row is one observed case.
 <V1> | <V2> | ... | <Vd>
 <obs_row_1_v1> | <obs_row_1_v2> | ... | <obs_row_1_vd>
 <obs_row_2_v1> | <obs_row_2_v2> | ... | <obs_row_2_vd>
 ...

--- INTERVENTIONAL DATA ---
 Each block corresponds to samples collected under a perfect intervention do(X = value).
 [Intervention: do(<X> = <value>)]
 <V1> | <V2> | ... | <Vd>
 <int_row_1_v1> | <int_row_1_v2> | ... | <int_row_1_vd>
 <int_row_2_v1> | <int_row_2_v2> | ... | <int_row_2_vd>
 ...

--- END OF DATA ---

--- OUTPUT INSTRUCTIONS ---
 Output exactly: <think>...</think><answer>...</answer>.
 Keep <think> concise (minimal necessary reasoning only).
 Inside <answer>, output exactly one JSON object with key "adjacency_matrix".
 - "adjacency_matrix": N x N 0/1 matrix in declared variable order.
 No extra text before, between, or after the two blocks.
 The JSON in <answer> must start with "{" and end with "}".

899 H.6 tabular Example

ROLE: You are an expert in causal discovery from observational and interventional data.
 TASK: Infer the directed causal graph over the variables.
 The following are empirical samples from a Bayesian network named cancer.
 ASSUMPTIONS:
 - The true graph is a DAG (no directed cycles).
 - Causal sufficiency holds (no unobserved confounders among these variables).
 - Interventions are perfect do-interventions (surgical): do(X=v) cuts all incoming edges into X.

--- VARIABLE ORDER (ORDER MATTERS) ---
 0: Pollution
 1: Smoker
 2: Cancer
 3: Xray
 4: Dyspnoea

--- OBSERVATIONAL DATA (no intervention) ---
 Each row is one observed case.
 Pollution | Smoker | Cancer | Xray | Dyspnoea
 low | False | False | negative | True
 low | False | False | negative | False
 low | False | False | positive | True
 low | False | False | positive | False
 high | False | False | positive | True
 low | False | False | positive | False
 high | False | False | negative | False
 low | False | False | negative | False
 low | False | False | negative | True
 low | True | False | positive | False

--- INTERVENTIONAL DATA ---

Each block corresponds to samples collected under a perfect intervention $do(X = \text{value})$.

```
[Intervention: do(Cancer = False)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | True | False | negative | True
low | True | False | negative | False
high | True | False | negative | True
```

```
[Intervention: do(Cancer = True)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | False | True | positive | False
low | True | True | positive | True
low | False | True | positive | True
```

```
[Intervention: do(Dyspnoea = False)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | False | False | negative | False
low | False | False | negative | False
low | True | False | negative | False
```

```
[Intervention: do(Dyspnoea = True)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | True | False | positive | True
high | True | False | negative | True
low | False | False | negative | True
```

```
[Intervention: do(Pollution = high)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
high | True | False | negative | True
high | False | False | positive | True
high | False | False | negative | False
```

```
[Intervention: do(Pollution = low)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | False | False | negative | False
low | True | False | negative | True
low | False | False | positive | False
```

```
[Intervention: do(Smoker = False)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | False | False | negative | False
high | False | False | negative | True
low | False | False | positive | False
```

```
[Intervention: do(Smoker = True)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | True | False | negative | False
low | True | False | negative | True
low | True | False | negative | False
```

```
[Intervention: do(Xray = negative)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
high | False | False | negative | False
low | True | False | negative | False
low | False | False | negative | False
```

```
[Intervention: do(Xray = positive)]
Pollution | Smoker | Cancer | Xray | Dyspnoea
low | False | False | positive | False
low | True | False | positive | False
low | True | False | positive | False
```

--- END OF DATA ---

--- OUTPUT INSTRUCTIONS ---

Output exactly: `<think>...</think><answer>...</answer>`.
 Keep `<think>` concise (minimal necessary reasoning only).
 Inside `<answer>`, output exactly one JSON object with key `"adjacency_matrix"`.
 - `"adjacency_matrix"`: $N \times N$ 0/1 matrix in declared variable order.
 No extra text before, between, or after the two blocks.
 The JSON in `<answer>` must start with `"{"` and end with `"}"`.

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Answer: [Yes]

Justification: The abstract and introduction clearly state the claims made, including the contributions made in the paper and important assumptions and limitations.

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